
EMCH 574 – Vibration and Waves Test 1

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Due date Not specified

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Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items.
- UG students are not required to do the G work, but they may do it for bonus points.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate.
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement [reference](#).
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga, and so on) [reference](#).
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit.
- Normalized plots are plots with the *y*-axis scaled between -1 and $+1$. A plotted variable is normalized by dividing it by its maximum absolute value.

Notations and Abbreviations

- BVP = boundary value problem.
- EOM = equation of motion.
- FBD = free body diagram.
- FRF = frequency response function.
- IVP = initial value problem.
- LHS = left hand side.
- NLM = Newton's law of motion.
- ODE = ordinary differential equation.
- PDE = partial differential equation.
- RHS = right hand side.

1 Pendulum

A.2 Calculate Pendulum Response to Initial Displacement

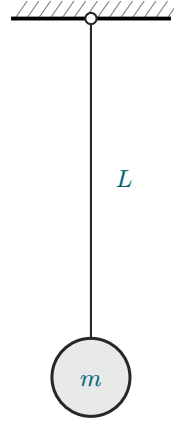


Figure 1: Pendulum of length L and mass m .

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L_{\text{nominal}} = 500 \text{ mm}$ and mass $m = 45 \text{ kg}$ as shown in Figure 1. When manufactured in series production, the pendulum lengths are not exact because of manufacturing tolerances. The lengths of the ten pendulum sets are

$$L = [502.013, 499.337, 500.472, 502.949, 502.216, \\ 496.635, 500.416, 498.255, 498.350, 499.358] \text{ mm}.$$

Do the following.

- (1) For the ten pendulum sets, calculate.
 - (a) Natural frequency ω_n in rad/s.
 - (b) Hz natural frequency f_n .
 - (c) Period of oscillation τ .
 - (d) Display in table format L, ω_n, f_n, τ .
- (2) Mean values: calculate and display the mean values of ω_n, f_n, τ .
- (3) Initial value problem (IVP). Assume initial displacement $u_0 = 4 \text{ mm}$ and zero initial velocity. Use the mean values of ω_n, f_n , and τ to complete items (a)–(c).
 - (a) Calculate and display the position of the mass $u(t_1)$ after $t_1 = 18 \text{ s}$.
 - (b) Calculate and display the duration $t_{N_{\text{osc}}}$ of N_{osc} oscillations and the corresponding position of the mass $u_{N_{\text{osc}}}$ for $N_{\text{osc}} = 15$.
 - (c) Plot the response of the system for N_{osc} oscillations and use a data tip to find $u(t_1)$ at t_1 .
- (4) Discuss your results.

Solution to A.2

Since the pendulum relation uses SI units, each length is first converted from millimetres to metres. For set j , the small-angle relation then gives

$$\omega_{n,j} = \sqrt{\frac{g}{L_j}} \quad (\text{A.2.1.a})$$

The frequency and period follow from $\omega_{n,j} = 2\pi f_{n,j}$ and $f_{n,j} = 1/\tau_j$, so

$$f_{n,j} = \frac{\omega_{n,j}}{2\pi} \quad (\text{A.2.1.b})$$

$$\tau_j = \frac{2\pi}{\omega_{n,j}} \quad (\text{A.2.1.c})$$

Using $g = 9.81 \text{ m/s}^2$ and the ten lengths gives the following values for A.2.1.d.

Set	L (mm)	ω_n (rad/s)	f_n (Hz)	τ (s)
1	502.013	4.42	0.704	1.421
2	499.337	4.43	0.705	1.418
3	500.472	4.43	0.705	1.419
4	502.949	4.42	0.703	1.423
5	502.216	4.42	0.703	1.422
6	496.635	4.44	0.707	1.414
7	500.416	4.43	0.705	1.419
8	498.255	4.44	0.706	1.416
9	498.350	4.44	0.706	1.416
10	499.358	4.43	0.705	1.418

The mean of each calculated column follows from $\bar{q} = \frac{1}{10} \sum_{j=1}^{10} q_j$. Therefore, the values required in A.2.2 are

$$\overline{\omega_n} = 4.43 \text{ rad/s} \quad (\text{A.2.2})$$

$$\overline{f_n} = 0.705 \text{ Hz} \quad (\text{A.2.2})$$

$$\overline{\tau} = 1.419 \text{ s} \quad (\text{A.2.2})$$

For A.2.3, the initial velocity is zero, so the undamped free response reduces to

$$u(t) = u_0 \cos(\overline{\omega_n} t) = 4 \cos(4.429470397t) \text{ mm} \quad (\text{A.2.3})$$

At $t_1 = 18$ s, substitution into the response gives

$$u(t_1) = 4 \cos(4.429470397 \times 18) \text{ mm} = -1.484 \text{ mm} \quad (\text{A.2.3.a})$$

For the requested fifteen-cycle duration, multiply the mean period by $N_{\text{osc}} = 15$. The duration and corresponding position are

$$t_{N_{\text{osc}}} = 15\bar{\tau} = 21.28 \text{ s} \quad (\text{A.2.3.b})$$

$$u_{N_{\text{osc}}} = u(15\bar{\tau}) = 4.00 \text{ mm} \quad (\text{A.2.3.b})$$

The mean of the individual periods is $\bar{\tau} = 1.4185009419$ s, whereas the period associated with the mean frequency is $2\pi/\bar{\omega}_n = 1.4184958345$ s. Averaging and taking a reciprocal do not commute. Following the prompt, the duration uses $15\bar{\tau}$; the difference from fifteen exact cycles of the mean-frequency model is only $76.6 \mu\text{s}$, and the endpoint still rounds to 4.00 mm.

Using the boxed A.2.3 expression over $0 \leq t \leq 15\bar{\tau}$ gives the fifteen-oscillation response below. Since the marker is placed at $t_1 = 18$ s, the plotted value is

$$u(18) = -1.484 \text{ mm} \quad (\text{A.2.3.c})$$

A2. Pendulum released from 4 mm

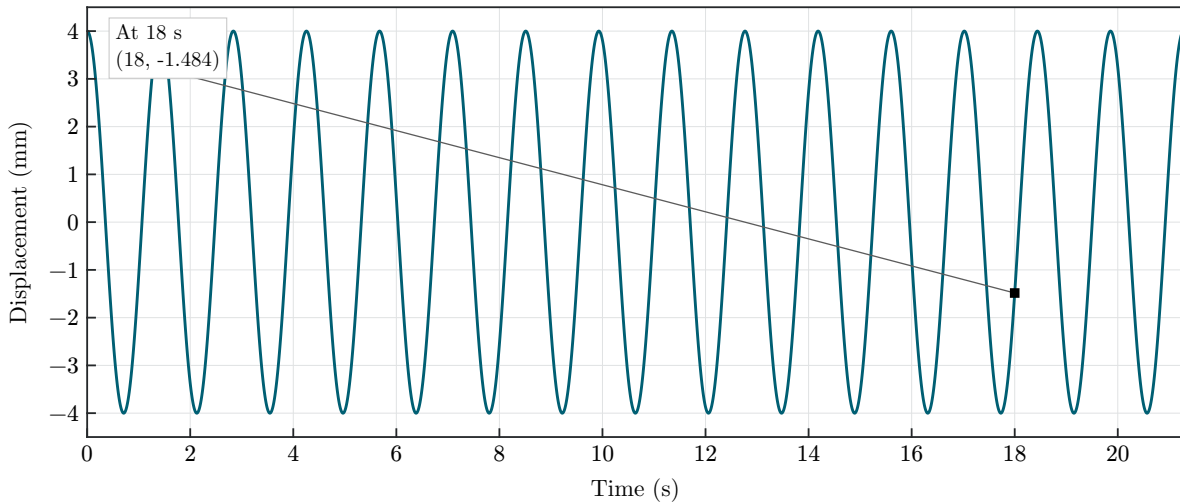


Figure 2: A.2.3.c. Pendulum response over fifteen oscillations with a data tip at $t = 18$ s.

MATLAB code. *Script used to reproduce the A.2 response figure.*

```

01 % A.2. Ten supplied lengths, their means, and displacement release.
02 cfg = hw01_settings();
03 g = 9.81; % m/s^2
04 Lmm = [502.013 499.337 500.472 502.949 502.216 ...
05        496.635 500.416 498.255 498.350 499.358];
06 L = Lmm/cfg.mmPerM; % m
07 u0 = 4/cfg.mmPerM; v0 = 0; % m, m/s
08 t1 = 18; Nosc = 15; % s, cycles
09 disp(table(g,u0,v0,t1,Nosc));
10 wnEach = sqrt(g./L);
11 fnEach = wnEach/(2*pi);
12 tauEach = 2*pi./wnEach;
13 disp(table(Lmm',wnEach',fnEach',tauEach', ...
14           'VariableNames',{'L_mm','wn_rad_s','fn_Hz','period_s'}));
15 wn = mean(wnEach); fn = mean(fnEach); tauMean = mean(tauEach);
16 periodOfMeanFrequency = 2*pi/wn;
17 disp(table(wn,fn,tauMean,periodOfMeanFrequency));
18 % The prompt requests the MEAN of the individual periods for duration.
19 tEnd = Nosc*tauMean;
20 t = hw01_grid(tEnd,periodOfMeanFrequency,t1);
21 u = hw01_free(t,wn,0,u0,v0);
22 u1 = hw01_free(t1,wn,0,u0,v0);
23 uEnd = hw01_free(tEnd,wn,0,u0,v0);
24 hw01_check_initial(wn,0,u0,v0);
25 disp(table(tEnd,u1*cfg.mmPerM,uEnd*cfg.mmPerM, ...
26           'VariableNames',{'duration_s','u18_mm','uEnd_mm'}));
27 hw01_plot('A2',t,u*cfg.mmPerM,t1,u1*cfg.mmPerM,[]);

```

A.3 Calculate Pendulum Response to Impact

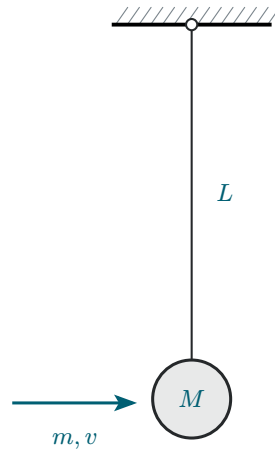


Figure 3: Impact excitation of a pendulum system.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L = 500 \text{ mm}$ and mass $M = 45 \text{ kg}$ as shown in Figure 3. The pendulum is initially at rest hanging vertically. A small projectile of mass $m = 11 \text{ g}$ and velocity $v = 850 \text{ m/s}$ hits the mass M and remains embedded into it. This impact imparts an initial velocity to the system that starts to oscillate.

Do the following.

- (1) Display input data.
- (2) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (3) Calculate the rad/sec natural frequency ω_n and period of oscillation τ of the pendulum with projectile embedded into it.
- (4) Calculate the response $u(t_1)$ after $t_1 = 18 \text{ s}$.
- (5) Plot the response of the system for $N_{\text{osc}} = 15$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response value calculated at item (4).
- (6) Discuss your results.

Solution to A.3

The input values required for A.3.1 are shown below. Since the response equations use SI units, the listed values are converted before substitution. The projectile mass is denoted by m_p so that it remains distinct from the pendulum mass M .

Quantity	Value	Unit
g	9.81	m/s ²
L	0.500	m
M	45.0	kg
m_p	0.0110	kg
v_p	850	m/s
t_1	18.0	s
N_{osc}	15	1

For the pendulum, the small-angle restoring motion gives the undamped equation

$$\ddot{u} + \frac{g}{L}u = 0.$$

Therefore, the natural frequency does not depend on the total mass, because the mass cancels from the ideal pendulum equation. The natural frequency and period are

$$\omega_n = \sqrt{\frac{g}{L}} = 4.42945 \text{ rad/s} \quad (\text{A.3.3})$$

$$\tau = \frac{2\pi}{\omega_n} = 1.41850 \text{ s} \quad (\text{A.3.3})$$

During the short impact, the projectile embeds in the pendulum mass, so linear momentum is conserved. Since the pendulum mass starts at rest, the common velocity immediately after impact satisfies

$$m_p v_p + M(0) = (M + m_p) \dot{u}_0.$$

Solving for the initial velocity gives

$$\dot{u}_0 = \frac{m_p v_p}{M + m_p} = 0.207727 \text{ m/s} \quad (\text{A.3.2})$$

The pendulum is still vertical immediately after impact, so $u_0 = 0$. Substitution into the undamped initial-value solution gives

$$u(t) = u_0 \cos(\omega_n t) + \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t).$$

Because $u_0 = 0$, the response becomes

$$u(t) = \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t) = 46.8968 \sin(4.42945t) \text{ mm} \quad (\text{A.3.4})$$

At $t_1 = 18 \text{ s}$, substitution into the response gives

$$u(t_1) = 46.8968 \sin(4.42945 \times 18) \text{ mm} = -43.5415 \text{ mm} \quad (\text{A.3.4})$$

Fifteen oscillations span the interval

$$t_{\text{end}} = N_{\text{osc}} \tau = 15(1.41850) = 21.2776 \text{ s} \quad (\text{A.3.5})$$

The response reaches its first positive maximum one quarter period after impact. The largest absolute response is therefore

$$|u|_{\text{max}} = \frac{\dot{u}_0}{\omega_n} = 46.8968 \text{ mm} \quad (\text{A.3.5.a})$$

The fifteen-oscillation plot includes data tips at the first maximum and at $t_1 = 18 \text{ s}$. The plotted readings are

$$u_{\text{max}} = 46.9 \text{ mm} \quad (\text{A.3.5.a})$$

$$u(18 \text{ s}) = -43.5 \text{ mm} \quad (\text{A.3.5.b})$$

A3. Response after an embedded-projectile impact

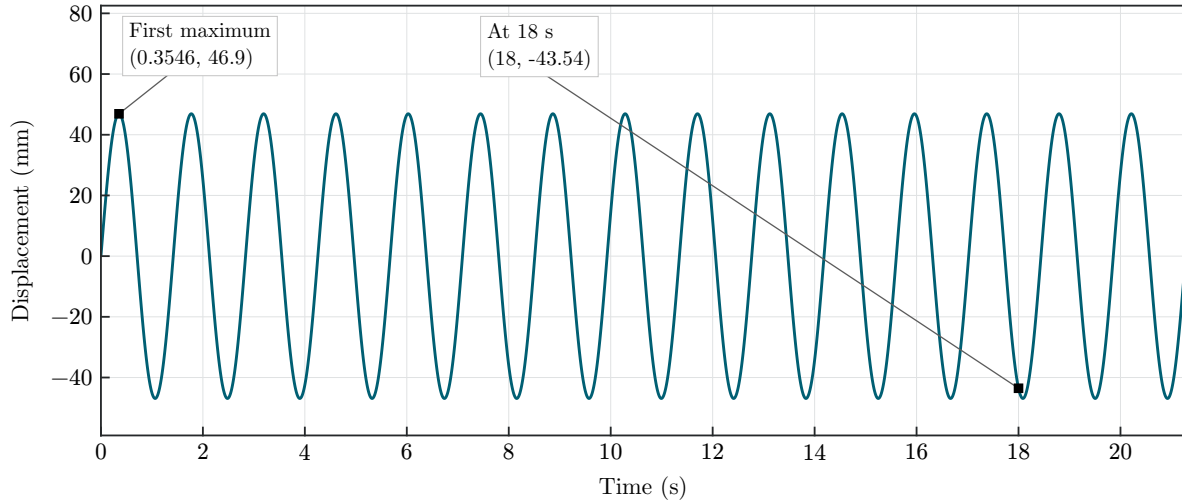


Figure 4: A.3.5.a-b. Impact response with data tips.

The projectile transfers its momentum to the combined mass, which produces a nonzero initial velocity but no initial displacement. The response consequently starts at zero, reaches its first maximum at $t = \tau/4 = 0.355$ s, and then repeats with constant amplitude because damping is neglected. The maximum displacement corresponds to an angular amplitude of approximately $u_{\max}/L = 0.0938$ rad = 5.37° , so the small-angle model is reasonable for this calculation. At 18 s, the pendulum is moving through the negative side of the motion rather than being at a maximum, which gives the plotted value of -43.5 mm.

MATLAB code. *Script used to plot the fifteen-oscillation response.*

```

01 % A.3. Perfectly inelastic projectile impact followed by pendulum motion.
02 cfg = hw01_settings();
03 g = 9.81; L = 0.500; M = 45;      % m/s^2, m, kg
04 mp = 0.011; vp = 850;            % kg, m/s
05 t1 = 18; Nosc = 15; u0 = 0;      % s, cycles, m
06 disp(table(g,L,M,mp,vp,t1,Nosc,u0));
07 totalMass = M+mp;
08 v0 = mp*vp/totalMass;
09 wn = sqrt(g/L); tau = 2*pi/wn;
10 amplitude = v0/wn; tPeak = tau/4;
11 eventT = [tPeak t1];
12 t = hw01_grid(Nosc*tau,tau,eventT);
13 u = hw01_free(t,wn,0,u0,v0);
14 eventU = hw01_free(eventT,wn,0,u0,v0);
15 hw01_check_initial(wn,0,u0,v0);
16 assert(abs(totalMass*v0-mp*vp)<cfg.checkTolerance);
17 disp(table(wn,tau,v0,amplitude*cfg.mmPerM, ...
18     eventU(2)*cfg.mmPerM,'VariableNames', ...
19     {'wn_rad_s','period_s','v0_m_s','amplitude_mm','u18_mm'}));
20 hw01_plot('A3',t,u*cfg.mmPerM,eventT,eventU*cfg.mmPerM,[]);

```

2 Spring-Mass-Damper Vibrating Systems

B.3 Calculate Spring-Mass-Damper Response to Initial Displacement

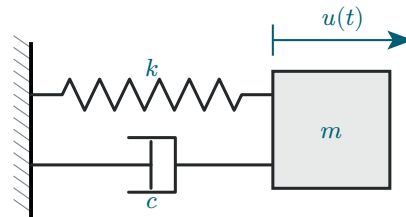


Figure 5: *Spring-mass-damper vibrating system.*

Consider a spring-mass vibrating system with $m = 5.3 \text{ kg}$, $k = 530 \text{ N/m}$, $c = 22 \text{ N/(m/s)}$ as shown in Figure 5. The vibrating system is given an initial displacement $u_0 = 4 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the natural frequency in rad/sec ω_n and in Hz f_n .
- (3) Calculate the critical damping c_{cr} .
- (4) Calculate the damping ratio ζ .
- (5) Calculate the damped frequency in rad/sec ω_d and in Hz f_d .
- (6) Calculate the damped period of oscillation τ_d .
- (7) Calculate the characteristic-equation roots s_1 , s_2 and the constants C_1 , C_2 .
- (8) Calculate the constants A , B and then the constants C , ϕ .
- (9) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.
- (10) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.
- (11) Plot the response of the system for $N_{osc} = 12$ oscillations and use datatips on the plot to find the following:
 - (a) The largest response.
 - (b) The response value calculated at item (9).
 - (c) The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (10).
- (12) Discuss your results.

Solution to B.3

The given system has mass $m = 5.3$ kg, stiffness $k = 530$ N/m, damping coefficient $c = 22$ N s/m, initial displacement $u_0 = 4$ mm, and initial velocity $\dot{u}_0 = 0$. Using $N_{\text{osc}} = 12$ sets the plotted interval to twelve damped oscillations.

Quantity	Value	Unit
m	5.3	kg
k	530	N/m
c	22.0	N s/m
u_0	4.00	mm
$\dot{u}_0$	0	mm/s
N_{osc}	12	–

The natural frequency follows from the stiffness-to-mass ratio. Converting it to hertz requires division by 2π .

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{530}{5.3}} = 10.000 \text{ rad/s}, \quad f_n = \frac{\omega_n}{2\pi} = 1.592 \text{ Hz}.$$

$$\boxed{\omega_n = 10.0 \text{ rad/s}, \quad f_n = 1.592 \text{ Hz}} \quad (\text{B.3.2})$$

Critical damping is the value that makes the characteristic-equation discriminant equal to zero. Therefore,

$$c_{\text{cr}} = 2\sqrt{mk} = 2\sqrt{(5.3)(530)} = 106.000 \text{ N s/m}.$$

$$\boxed{c_{\text{cr}} = 106.0 \text{ N s/m}} \quad (\text{B.3.3})$$

The damping ratio compares the actual damping with the critical value. Since $c < c_{\text{cr}}$, the system is underdamped.

$$\zeta = \frac{c}{c_{\text{cr}}} = \frac{22}{106.000} = 0.207547.$$

$$\boxed{\zeta = 0.2075 \quad (\text{underdamped})} \quad (\text{B.3.4})$$

For an underdamped system, the oscillatory part of the characteristic roots has frequency $\omega_d = \omega_n \sqrt{1 - \zeta^2}$. Converting this frequency to hertz gives

$$\omega_d = 10.000\sqrt{1 - (0.207547)^2} = 9.782 \text{ rad/s}, \quad f_d = \frac{\omega_d}{2\pi} = 1.557 \text{ Hz}.$$

$$\boxed{\omega_d = 9.78 \text{ rad/s}, \quad f_d = 1.557 \text{ Hz}} \quad (\text{B.3.5})$$

One damped oscillation lasts one damped period. Thus,

$$\tau_d = \frac{2\pi}{\omega_d} = \frac{2\pi}{9.782} = 0.6423 \text{ s}.$$

$$\boxed{\tau_d = 0.642 \text{ s}} \quad (\text{B.3.6})$$

The real part of each root is the exponential decay rate $-\zeta\omega_n = -c/(2m) = -2.075472 \text{ s}^{-1}$. Combining this decay rate with the damped frequency gives

$$s_{1,2} = -\zeta\omega_n \pm i\omega_d = -2.075472 \pm i 9.7823 \text{ s}^{-1}.$$

For the exponential form $u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$, the real trigonometric coefficients satisfy $A = C_1 + C_2$ and $B = i(C_1 - C_2)$. The initial conditions determine $A = u_0$ and $B = \zeta\omega_n u_0 / \omega_d$.

$$C_1 = \frac{A - iB}{2} = 2.000 - 0.4243i \text{ mm}, \quad C_2 = \frac{A + iB}{2} = 2.000 + 0.4243i \text{ mm}.$$

$$\boxed{\begin{array}{ll} s_1 = -2.0755 + i 9.7823 \text{ s}^{-1}, & s_2 = -2.0755 - i 9.7823 \text{ s}^{-1}, \\ C_1 = 2.000 - 0.4243i \text{ mm}, & C_2 = 2.000 + 0.4243i \text{ mm} \end{array}} \quad (\text{B.3.7})$$

The zero initial velocity condition fixes the coefficient of the sine term. Substituting the numerical values gives

$$A = u_0 = 4.00 \text{ mm}, \quad B = \frac{\zeta\omega_n u_0}{\omega_d} = 0.8487 \text{ mm}.$$

Writing the harmonic part as a single sine function gives $A = C \sin \phi$ and $B = C \cos \phi$. Therefore,

$$C = \sqrt{A^2 + B^2} = 4.089 \text{ mm}, \quad \phi = \text{atan2}(A, B) = 1.362 \text{ rad}.$$

The displacement response is consequently

$$u(t) = 4.089e^{-2.0755t} \sin(9.7823t + 1.362) \text{ mm.}$$

$$\boxed{\begin{array}{ll} A = 4.00 \text{ mm}, & B = 0.849 \text{ mm}, \\ C = 4.09 \text{ mm}, & \phi = 1.362 \text{ rad} \end{array}} \quad (\text{B.3.8})$$

At one damped period, the sine and cosine terms return to their initial values. The oscillatory factor is therefore one, while the exponential factor has decayed for one period.

$$t_1 = \tau_d = 0.6423 \text{ s}, \quad u_1 = u(t_1) = u_0 e^{-\zeta \omega_n t_1} = 1.055 \text{ mm.}$$

$$\boxed{t_1 = 0.642 \text{ s}, \quad u_1 = 1.055 \text{ mm}} \quad (\text{B.3.9})$$

The displacement magnitude defining the 2% band is

$$u_{0.02} = 0.02u_0 = 0.02(4.0) = 0.0800 \text{ mm.}$$

$$\boxed{u_{0.02} = 0.0800 \text{ mm}} \quad (\text{B.3.10})$$

The largest response occurs at release because the initial displacement is 4.0 mm and the envelope decreases thereafter. The one-period response is the value calculated above. To find the permanent settling time, solve $u(t) = -u_{0.02}$ on the rising side of the last negative lobe, as the displacement returns toward zero. The numerical crossing is $t_s = 1.7110 \text{ s}$. All later extrema have smaller magnitude, so the response remains inside the 2% band after this crossing.

$$\boxed{|u|_{\max} = 4.00 \text{ mm} \quad \text{at } t = 0} \quad (\text{B.3.11.a})$$

$$\boxed{u(t_1) = 1.055 \text{ mm} \quad \text{at } t_1 = 0.642 \text{ s}} \quad (\text{B.3.11.b})$$

$$\boxed{t_s = 1.71 \text{ s}, \quad u(t_s) = -0.0800 \text{ mm}} \quad (\text{B.3.11.c})$$

B3. Displacement release

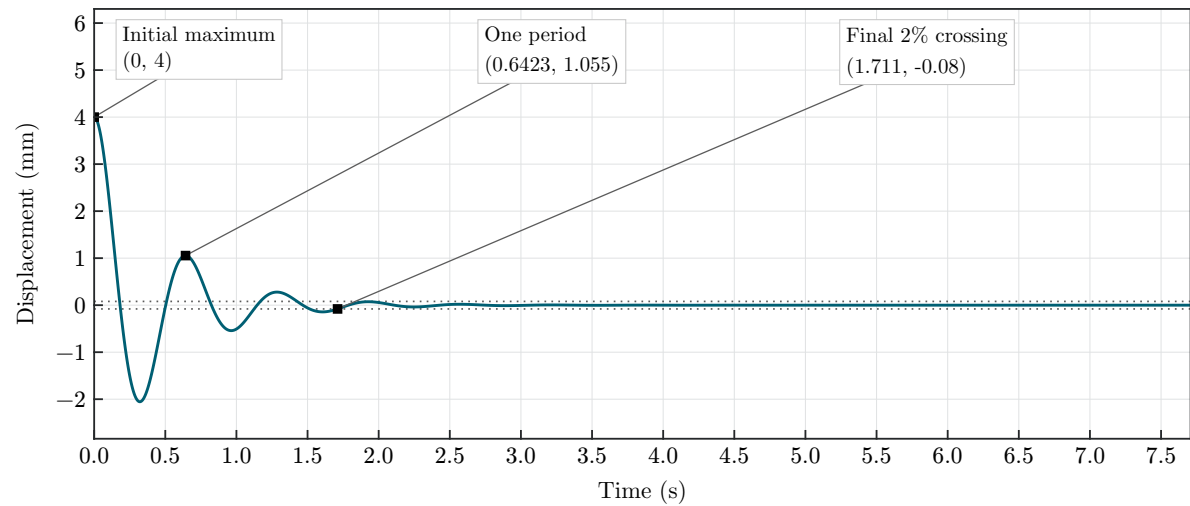


Figure 6: B.3.11.a-b. Twelve-cycle response with the initial maximum, one-period response, and final 2% crossing.

B3. Final crossing of the 2% band

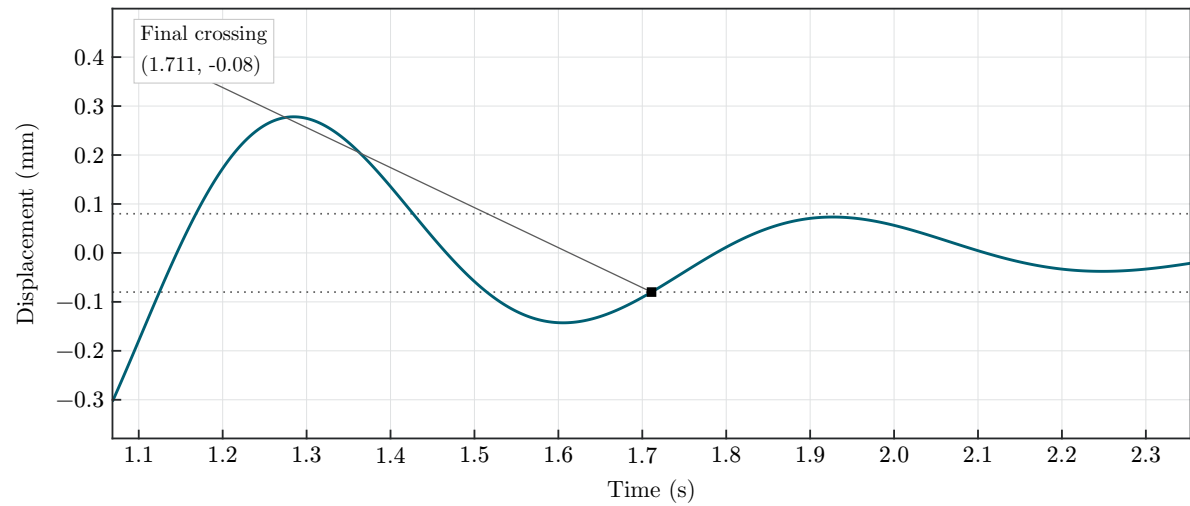


Figure 7: B.3.11.c. Detail of the final 2% crossing of the settling band.

MATLAB code. *MATLAB code used to reproduce the B.3.11 response plot.*

```

01 % B.3. Damped displacement release and permanent 2 percent settling.
02 cfg = hw01_settings();
03 m = 5.3; k = 530; c = 22;          % kg, N/m, N s/m
04 u0 = 4/cfg.mmPerM; v0 = 0;       % m, m/s
05 Nosc = 12;
06 disp(table(m,k,c,u0,v0,Nosc));
07 wn = sqrt(k/m); cCritical = 2*sqrt(m*k);
08 zeta = c/cCritical; wd = wn*sqrt(1-zeta^2);
09 tau = 2*pi/wd;
10 [ts,us,tEnvelope] = hw01_settling(wn,zeta,u0,cfg.settlingFraction);
11 eventT = [0 tau ts];
12 eventU = hw01_free(eventT,wn,zeta,u0,v0);
13 t = hw01_grid(Nosc*tau,tau,eventT);
14 u = hw01_free(t,wn,zeta,u0,v0);
15 hw01_check_initial(wn,zeta,u0,v0);
16 disp(table(wn,cCritical,zeta,wd,tau,ts,tEnvelope));
17 alpha=zeta*wn; s1=-alpha+1i*wd; s2=conj(s1);
18 A=u0; B=(v0+alpha*u0)/wd;
19 C1=(A-1i*B)/2; C2=conj(C1);
20 amplitude=hypot(A,B); phase=atan2(A,B);
21 disp(table(s1,s2,C1,C2,A,B,amplitude,phase));
22 disp(table(eventT',eventU'*cfg.mmPerM, ...
23     'VariableNames',{'time_s','displacement_mm'}));
24 band = cfg.settlingFraction*abs(u0)*cfg.mmPerM;
25 hw01_plot('B3',t,u*cfg.mmPerM,eventT,eventU*cfg.mmPerM,band);
26 % A two-period detail makes the final crossing visible.
27 detailStart = max(0,ts-tau); detailEnd = ts+tau;
28 td = unique([linspace(detailStart,detailEnd, ...
29     2*cfg.samplesPerCycle+1),ts]);
30 ud = hw01_free(td,wn,zeta,u0,v0);
31 hw01_plot('B3_settling',td,ud*cfg.mmPerM,ts,us*cfg.mmPerM,band);

```

The plot shows an underdamped response whose oscillation frequency remains constant while its amplitude decreases exponentially. The initial condition produces the largest displacement at release, and the displacement after one period is smaller by the factor $e^{-\zeta\omega_n\tau_d}$. The response first enters the 2% band before 1.71 s, but it does not remain there until the final crossing at $t_s = 1.71$ s. The exponential envelope gives the conservative bound $t_{\text{envelope}} = 1.90$ s, which is later because the response is not at the envelope maximum at the exact crossing.

(B.3.12)

B.4 Calculate Truck Suspension Response

Consider a pickup truck wheel suspension consisting of a leaf spring and an oil damper. The truck and its suspension can be modeled, in first approximation, as a k - m - c spring-mass-damper system as shown in Figure 8.

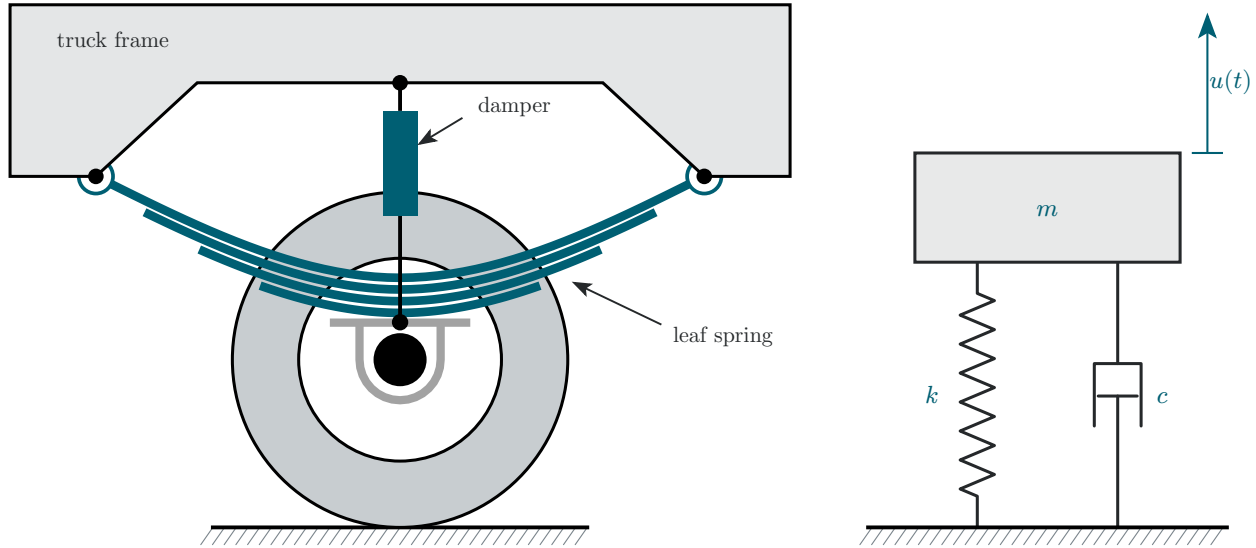


Figure 8: Truck suspension consisting of a leaf spring and an oil damper. The wheel contact is treated as the road datum in the one-degree-of-freedom idealization.

The leaf spring can be approximated by a uniform beam of length $L = 1200 \text{ mm}$ and cross-section dimensions $h = 4 \text{ mm}$ and $b = 40 \text{ mm}$, made of steel with $E = 200 \text{ GPa}$. The truck mass apportioned to the wheel can be estimated as $m = 520 \text{ kg}$. Assume initial displacement $u_0 = 105 \text{ mm}$ and zero initial velocity $\dot{u}_0 = 0$.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I of the beam cross section.
- (3) Calculate the flexural stiffness EI of the beam.
- (4) Calculate the spring constant k of the system.
- (5) Calculate the natural frequency in rad/sec ω_n and in Hz f_n .
- (6) Calculate the critical damping c_{cr} .
- (7) Assume a 25% damping ratio, i.e., $\zeta = 0.25$, and calculate the physical damping c . Print ζ and c .
- (8) Calculate the damped frequency in rad/sec ω_d and in Hz f_d .
- (9) Calculate the damped period of oscillation τ_d .
- (10) Calculate the characteristic-equation roots s_1, s_2 and the constants C_1, C_2 .
- (11) Calculate the constants A, B and then the constants C, ϕ .

-
- (12) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.
- (13) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.
- (14) Plot the response of the system for $N_{\text{osc}} = 17$ oscillations and use datatips on the plot to find the following:
- (a) The largest response.
 - (b) The response value calculated at item (12).
 - (c) The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (13).
- (15) Discuss your results.
- (16) Graduate work. Assume the damper is worn out such that ζ is reduced to $\zeta = 10\%$.
- (a) Repeat the calculation of items (6) through (15).
 - (b) Compare and discuss the two results.

Solution to B.4

The beam calculation uses SI units, while the response plots report displacement in millimetres. I idealize the leaf spring as a simply supported beam with a concentrated load at its center because this matches the two spring-eye supports and the center displacement shown in the model. Therefore, the corresponding center deflection is

$$\delta = \frac{PL^3}{48EI}, \quad k = \frac{P}{\delta} = \frac{48EI}{L^3}.$$

This support condition is important because a cantilever model would produce a different stiffness.

B.4.1. Input data.

Quantity	Value	Unit
L	1.200	m
b	0.040	m
h	0.004	m
E	200	GPa
m	520	kg
u_0	105	mm
$\dot{u}_0$	0	mm/s
ζ_H	0.25	1
ζ_W	0.10	1
g	9.81	m/s ²

For vertical bending about the horizontal centroidal axis, the section dimension perpendicular to that axis is h , so h is cubed in the second moment of area.

$$I = \frac{bh^3}{12} = \frac{(0.040)(0.004)^3}{12} = 2.13333 \times 10^{-10} \text{ m}^4 = 213.333 \text{ mm}^4.$$

$$\boxed{I = 2.13333 \times 10^{-10} \text{ m}^4 = 213.333 \text{ mm}^4} \quad (\text{B.4.2})$$

Multiplying the section property by the elastic modulus gives the flexural rigidity.

$$EI = (200 \times 10^9)(2.13333 \times 10^{-10}) = 42.6667 \text{ N m}^2.$$

$$\boxed{EI = 42.6667 \text{ N m}^2} \quad (\text{B.4.3})$$

Substitution into the simply supported center-load relation gives the equivalent spring constant.

$$k = \frac{48EI}{L^3} = \frac{48(42.6667)}{(1.200)^3} = 1185.19 \text{ N/m}.$$

$$\boxed{k = 1.185 \times 10^3 \text{ N/m}} \quad (\text{B.4.4})$$

The undamped natural frequency follows from the mass and equivalent stiffness. Its cyclic frequency is found by dividing by 2π .

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{1185.19}{520}} = 1.50970 \text{ rad/s}, \quad f_n = \frac{\omega_n}{2\pi} = 0.240277 \text{ Hz}.$$

$$\boxed{\omega_n = 1.51 \text{ rad/s}, \quad f_n = 0.2403 \text{ Hz}} \quad (\text{B.4.5})$$

Critical damping makes the characteristic-equation roots coincide. Therefore,

$$c_{\text{cr}} = 2\sqrt{mk} = 2\sqrt{(520)(1185.19)} = 1570.09 \text{ N s/m}.$$

$$\boxed{c_{\text{cr}} = 1.570 \times 10^3 \text{ N s/m}} \quad (\text{B.4.6})$$

For the original damper, the specified damping ratio is $\zeta_{\text{H}} = 0.25$. The physical damping is the damping ratio multiplied by the critical value.

$$c_{\text{H}} = \zeta_{\text{H}} c_{\text{cr}} = (0.25)(1570.09) = 392.523 \text{ N s/m}.$$

$$\boxed{\zeta_{\text{H}} = 0.250, \quad c_{\text{H}} = 392.5 \text{ N s/m}} \quad (\text{B.4.7})$$

Because $\zeta_{\text{H}} < 1$, the original system is underdamped. Its damped frequency and cyclic frequency are

$$\omega_{d,\text{H}} = \omega_n \sqrt{1 - \zeta_{\text{H}}^2} = 1.50970 \sqrt{1 - 0.25^2} = 1.46176 \text{ rad/s}, \quad f_{d,\text{H}} = \frac{\omega_{d,\text{H}}}{2\pi} = 0.232647 \text{ Hz}.$$

$$\boxed{\omega_{d,\text{H}} = 1.46 \text{ rad/s}, \quad f_{d,\text{H}} = 0.2326 \text{ Hz}} \quad (\text{B.4.8})$$

The time for one damped oscillation is the damped period.

$$\tau_{d,H} = \frac{2\pi}{\omega_{d,H}} = \frac{2\pi}{1.46176} = 4.29836 \text{ s.}$$

$$\boxed{\tau_{d,H} = 4.30 \text{ s}} \quad (\text{B.4.9})$$

Defining the decay rate as $\alpha_H = \zeta_H \omega_n = 0.377426 \text{ s}^{-1}$ gives the characteristic roots

$$s_{1,2} = -\alpha_H \pm i\omega_{d,H} = -0.377426 \pm i 1.46176 \text{ s}^{-1}.$$

For the zero-velocity initial condition, the exponential constants are obtained from the trigonometric coefficients $A = u_0$ and $B = \alpha_H u_0 / \omega_{d,H}$. Since $A = C_1 + C_2$ and $B = i(C_1 - C_2)$,

$$C_1 = \frac{A - iB}{2} = 52.500 - 13.555i \text{ mm}, \quad C_2 = \frac{A + iB}{2} = 52.500 + 13.555i \text{ mm}.$$

$$\boxed{\begin{aligned} s_1 &= -0.3774 + i 1.4618 \text{ s}^{-1}, & s_2 &= -0.3774 - i 1.4618 \text{ s}^{-1}, \\ C_1 &= 52.50 - 13.56i \text{ mm}, & C_2 &= 52.50 + 13.56i \text{ mm} \end{aligned}} \quad (\text{B.4.10})$$

The real trigonometric coefficients are

$$A = u_0 = 105.0 \text{ mm}, \quad B = \frac{\alpha_H u_0}{\omega_{d,H}} = 27.1109 \text{ mm}.$$

Writing the harmonic part as a single sine function gives $A = C \sin \phi$ and $B = C \cos \phi$. Therefore,

$$C = \sqrt{A^2 + B^2} = 108.444 \text{ mm}, \quad \phi = \text{atan2}(A, B) = 1.31812 \text{ rad}.$$

The displacement response for the original damper is consequently

$$u_H(t) = 108.444e^{-0.377426t} \sin(1.46176t + 1.31812) \text{ mm}.$$

$$\boxed{\begin{aligned} A &= 105.0 \text{ mm}, & B &= 27.11 \text{ mm}, \\ C &= 108.4 \text{ mm}, & \phi &= 1.318 \text{ rad} \end{aligned}} \quad (\text{B.4.11})$$

At one damped period, the sine and cosine terms return to their initial values. The displacement is therefore the initial displacement multiplied by the decay factor for one period.

$$t_{1,H} = \tau_{d,H} = 4.29836 \text{ s}, \quad u_{1,H} = u_H(t_{1,H}) = u_0 e^{-\alpha_H t_{1,H}} = 20.731 \text{ mm}.$$

$$\boxed{t_{1,H} = 4.30 \text{ s}, \quad u_{1,H} = 20.73 \text{ mm}} \quad (\text{B.4.12})$$

The displacement magnitude defining the 2% band is

$$u_{0.02} = 0.02u_0 = 0.02(105) = 2.10 \text{ mm}.$$

$$\boxed{u_{0.02} = 2.10 \text{ mm}} \quad (\text{B.4.13})$$

The largest response is the initial displacement, so $|u|_{\max} = 105 \text{ mm}$ at $t = 0$. The last extremum with magnitude greater than $u_{0.02}$ occurs at approximately $t = 8.597 \text{ s}$, where $u = 4.093 \text{ mm}$. Solving the descending branch for $u(t) = u_{0.02}$ gives the final entry into the 2% band at $t_s = 9.351 \text{ s}$. All later extrema have magnitude below 2.10 mm .

$$\boxed{|u|_{\max} = 105.0 \text{ mm} \quad \text{at } t = 0} \quad (\text{B.4.14.a})$$

$$\boxed{u(t_{1,H}) = 20.73 \text{ mm} \quad \text{at } t_{1,H} = 4.30 \text{ s}} \quad (\text{B.4.14.b})$$

$$\boxed{t_{s,H} = 9.35 \text{ s}, \quad u_H(t_{s,H}) = +2.10 \text{ mm}} \quad (\text{B.4.14.c})$$

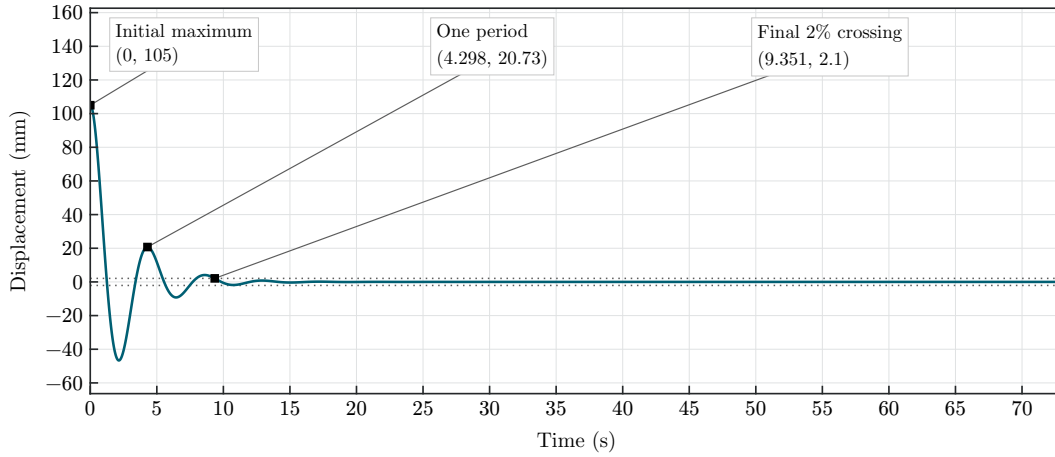
B4 healthy. Displacement release

Figure 9: *B.4.14.a-b-c. Original damper response for seventeen oscillations.*

B.4.15. Discussion.

The original damper produces a decaying underdamped response. The damped period is 4.30 s, and the displacement after one period is reduced from 105 mm to 20.73 mm. The response remains oscillatory, but its extrema decrease because the envelope is $Ce^{-\alpha_H t}$. The final permanent entry into the 2% band occurs at 9.35 s.

The calculated static displacement under the apportioned weight is

$$\delta_{\text{static}} = \frac{mg}{k} = \frac{(520)(9.81)}{1185.19} = 4.304 \text{ m}.$$

This is larger than the beam span and much larger than the prescribed 105 mm initial displacement. The value indicates that the uniform single-beam model is a formal linear approximation. A physical suspension would require the actual leaf-spring stack, support geometry, preload, and available travel.

For the worn damper, the stiffness and mass do not change. Consequently, I , EI , k , ω_n , f_n , and c_{cr} remain unchanged. Using $\zeta_W = 0.10$ therefore gives the new damping and decay rate

$$c_W = \zeta_W c_{\text{cr}} = 157.009 \text{ N s/m}, \quad \alpha_W = \zeta_W \omega_n = 0.150970 \text{ s}^{-1}.$$

The smaller damping ratio gives

$$\omega_{d,W} = \omega_n \sqrt{1 - \zeta_W^2} = 1.50214 \text{ rad/s}, \quad f_{d,W} = 0.239072 \text{ Hz},$$

$$\tau_{d,W} = \frac{2\pi}{\omega_{d,W}} = 4.18284 \text{ s}.$$

The worn-damper roots and exponential constants are

$$s_{1,2}^W = -0.150970 \pm i 1.50214 \text{ s}^{-1}, \quad C_1^W = 52.500 - 5.276i \text{ mm}, \quad C_2^W = 52.500 + 5.276i \text{ mm}.$$

The corresponding real coefficients are

$$A_W = 105.0 \text{ mm}, \quad B_W = 10.553 \text{ mm}, \quad C_W = 105.529 \text{ mm}, \quad \phi_W = 1.47063 \text{ rad}.$$

Thus, the worn-damper response is

$$u_W(t) = 105.529e^{-0.150970t} \sin(1.50214t + 1.47063) \text{ mm}.$$

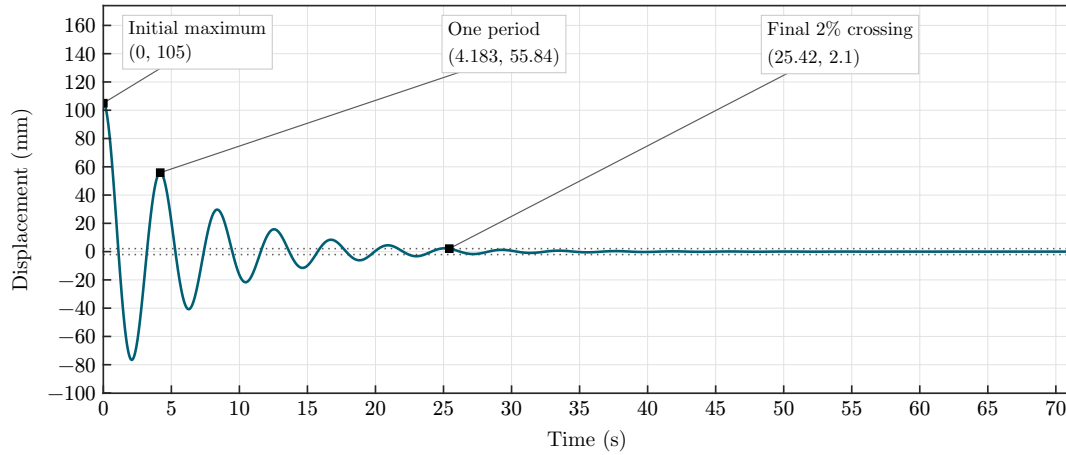
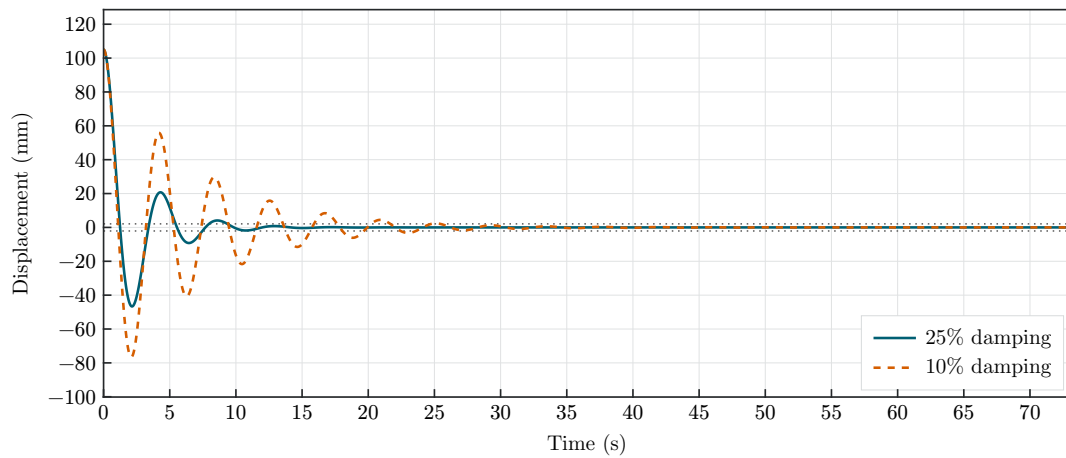
The response after one period and the final permanent 2% crossing are

$$t_{1,W} = \tau_{d,W} = 4.18284 \text{ s}, \quad u_{1,W} = u_0 e^{-\alpha_W t_{1,W}} = 55.839 \text{ mm},$$

$$t_{s,W} = 25.424 \text{ s}, \quad u_W(t_{s,W}) = +u_{0.02} = 2.10 \text{ mm}.$$

$c_{cr} = 1570.09 \text{ N s/m},$	$c_W = 157.01 \text{ N s/m},$	(B.4.16.a)
$\omega_{d,W} = 1.502 \text{ rad/s},$	$f_{d,W} = 0.2391 \text{ Hz},$	
$\tau_{d,W} = 4.183 \text{ s},$	$s_{1,2}^W = -0.1510 \pm i 1.502 \text{ s}^{-1},$	
$C_1^W = 52.50 - 5.276i \text{ mm},$	$C_2^W = 52.50 + 5.276i \text{ mm},$	
$A_W = 105.0 \text{ mm},$	$B_W = 10.55 \text{ mm},$	
$C_W = 105.5 \text{ mm},$	$\phi_W = 1.471 \text{ rad},$	
$t_{1,W} = 4.183 \text{ s},$	$u_{1,W} = 55.84 \text{ mm}$	

B.4.16.b. Comparison. The worn damper has a smaller decay rate, so its oscillations persist longer. Its damped frequency is slightly higher and its period is slightly shorter because reducing damping moves ω_d closer to ω_n . The one-period displacement increases from 20.73 mm to 55.84 mm, and the permanent 2% settling time increases from 9.35 s to 25.42 s. Both cases have the same initial maximum because they have the same initial displacement.

B4 worn. Displacement release**Figure 10:** *B.4.16.a. Worn damper response for seventeen oscillations.***B4. Worn damping increases the duration of motion****Figure 11:** *B.4.16.b. Comparison of the original and worn-damper responses.*

MATLAB code. *MATLAB code used to reproduce the B.4.14 and B.4.16 response plots.*

```

01 % B.4. Simply supported leaf spring with a central concentrated load.
02 cfg = hw01_settings();
03 L = 1.200; b = .040; h = .004; % m
04 E = 200e9; m = 520; g = 9.81; % Pa, kg, m/s^2
05 u0 = 105/cfg.mmPerM; v0 = 0; % m, m/s
06 Nosc = 17; zeta = [.25 .10];
07 disp(table(L,b,h,E,m,g,u0,v0,Nosc)); disp(zeta);
08 I = b*h^3/12; % Rectangular second moment, m^4.
09 k = 48*E*I/L^3; % Center-load beam stiffness, N/m.
10 wn = sqrt(k/m); cCritical = 2*m*wn;
11 staticDeflection = m*g/k;
12 disp(table(I,k,wn,cCritical,staticDeflection));
13 names = {'B4_healthy', 'B4_worn'};
14 periods = 2*pi./(wn*sqrt(1-zeta.^2));
15 band = cfg.settlingFraction*abs(u0)*cfg.mmPerM;
16 for j=1:numel(zeta)
17     tau = periods(j); damping = zeta(j)*cCritical;
18     [ts,us,tEnvelope] = hw01_settling( ...
19         wn,zeta(j),u0,cfg.settlingFraction);
20     eventT = [0 tau ts];
21     eventU = hw01_free(eventT,wn,zeta(j),u0,v0);
22     t = hw01_grid(Nosc*tau,tau,eventT);
23     u = hw01_free(t,wn,zeta(j),u0,v0);
24     hw01_check_initial(wn,zeta(j),u0,v0);
25     wd=2*pi/tau; alpha=zeta(j)*wn;
26     s1=-alpha+1i*wd; s2=conj(s1);
27     A=u0; B=(v0+alpha*u0)/wd;
28     C1=(A-1i*B)/2; C2=conj(C1);
29     amplitude=hypot(A,B); phase=atan2(A,B);
30     disp(table(s1,s2,C1,C2,A,B,amplitude,phase));
31     disp(table(zeta(j),damping,tau,ts,us*cfg.mmPerM,tEnvelope, ...
32         'VariableNames',{'zeta','c_Ns_m','period_s', ...
33         'settling_s','crossing_mm','envelope_s'}));
34     hw01_plot(names{j},t,u*cfg.mmPerM, ...
35         eventT,eventU*cfg.mmPerM,band);
36 end
37 t = hw01_grid(Nosc*max(periods),min(periods),[]);
38 u = zeros(numel(zeta),numel(t));
39 for j=1:numel(zeta)
40     u(j,:) = hw01_free(t,wn,zeta(j),u0,v0);
41 end
42 hw01_plot('B4_compare',t,u*cfg.mmPerM,[],[],band);

```

B.5 Overdamped Spring-Mass-Damper Systems

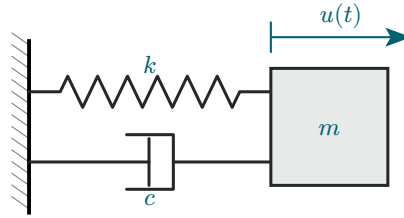


Figure 12: Spring-mass-damper system used for the overdamped and critically damped responses.

Consider a spring-mass-damper system with natural frequency $f_n = 5 \text{ Hz}$ and various values of the damping ratio ζ , as shown in Figure 12. The system is given an initial displacement $u_0 = 3 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the rad/s frequency ω_n .
- (3) Assume overdamped conditions $\zeta > 1$ and develop the solution as follows:
 - (a) Write the expressions of the roots s_1, s_2 in terms of ω_n and ζ .
 - (b) Write the solution $u(t)$ in terms of s_1, s_2 and constants C_1, C_2 .
 - (c) Write the expression of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
 - (d) Take $\zeta = 2.5$ and calculate the numerical values of s_1, s_2, C_1 , and C_2 .
 - (e) Plot the response $u(t)$ for $t = 0, \dots, 3 \text{ s}$.
- (4) Assume critically damped conditions $\zeta = 1$ and complete items (a)–(e).
 - (a) Write the expressions of the two partial solutions $u_1(t), u_2(t)$ in terms of ω_n .
 - (b) Write the expression of the complete solution in terms of ω_n and C_1, C_2 .
 - (c) Write the expressions of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
 - (d) Calculate the numerical values of C_1, C_2 .
 - (e) Plot the response $u(t)$ for $t = 0, \dots, 3 \text{ s}$.
- (5) Graduate work. Nothing here.
- (6) Graduate work. Some race-car manufacturers design for critical damping, whereas others use an overdamped design. Discuss the advantages and disadvantages of the two options.

Solution to B.5

The given initial conditions are $u(0) = u_0 = 3$ mm and $\dot{u}(0) = \dot{u}_0 = 0$. The natural frequency is specified by $f_n = 5$ Hz.

B.5.1. Input data.

Quantity	Value	Unit
f_n	5	Hz
u_0	3.0	mm
$\dot{u}_0$	0	mm/s
t	0–3	s

B.5.2. Natural frequency.

The frequency in hertz is converted to angular frequency by multiplying by 2π .

$$\omega_n = 2\pi f_n = 2\pi(5) = 31.4159 \text{ rad/s.}$$

$$\boxed{\omega_n = 31.42 \text{ rad/s}} \quad (\text{B.5.2})$$

B.5.3. Overdamped response with $\zeta = 2.5$.

For free vibration, the characteristic equation is

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0.$$

(a) Roots. Since $\zeta > 1$, the discriminant is positive and the two roots are real. Letting s_1 denote the faster root and s_2 the slower root gives

$$s_1 = -\left(\zeta + \sqrt{\zeta^2 - 1}\right)\omega_n, \quad s_2 = -\left(\zeta - \sqrt{\zeta^2 - 1}\right)\omega_n.$$

$$\boxed{s_1 = -\left(\zeta + \sqrt{\zeta^2 - 1}\right)\omega_n, \quad s_2 = -\left(\zeta - \sqrt{\zeta^2 - 1}\right)\omega_n} \quad (\text{B.5.3.a})$$

(b) General solution. Each real root produces an exponential partial solution. Their linear combination is

$$\boxed{u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}} \quad (\text{B.5.3.b})$$

(c) Initial-condition constants. Evaluating the response at $t = 0$ gives $C_1 + C_2 = u_0$. Differentiating first and then evaluating at $t = 0$ gives $s_1 C_1 + s_2 C_2 = \dot{u}_0$. Solving these two equations yields

$$\boxed{C_1 = \frac{\dot{u}_0 - s_2 u_0}{s_1 - s_2}, \quad C_2 = \frac{s_1 u_0 - \dot{u}_0}{s_1 - s_2}} \quad (\text{B.5.3.c})$$

(d) **Numerical values.** For $\zeta = 2.5$,

$$\sqrt{\zeta^2 - 1} = \sqrt{2.5^2 - 1} = 2.29129.$$

Substitution into the root expressions gives

$$\begin{aligned} s_1 &= -(2.5 + 2.29129)(31.4159) = -150.523 \text{ s}^{-1}, \\ s_2 &= -(2.5 - 2.29129)(31.4159) = -6.55689 \text{ s}^{-1}. \end{aligned}$$

Using $u_0 = 3 \text{ mm}$ and $\dot{u}_0 = 0$ in the constant expressions gives

$$C_1 = \frac{0 - (-6.55689)(3)}{-150.523 - (-6.55689)} = -0.13663 \text{ mm},$$

$$C_2 = \frac{(-150.523)(3) - 0}{-150.523 - (-6.55689)} = 3.13663 \text{ mm}.$$

$$\boxed{\begin{aligned} s_1 &= -150.523 \text{ s}^{-1}, & s_2 &= -6.55689 \text{ s}^{-1}, \\ C_1 &= -0.13663 \text{ mm}, & C_2 &= 3.13663 \text{ mm} \end{aligned}} \quad (\text{B.5.3.d})$$

Therefore, the overdamped displacement is

$$u_{\text{OD}}(t) = -0.13663e^{-150.523t} + 3.13663e^{-6.55689t} \text{ mm}.$$

(e) **Overdamped plot.** The response is plotted over the required interval $0 \leq t \leq 3 \text{ s}$.

B5. Critical and overdamped displacement release

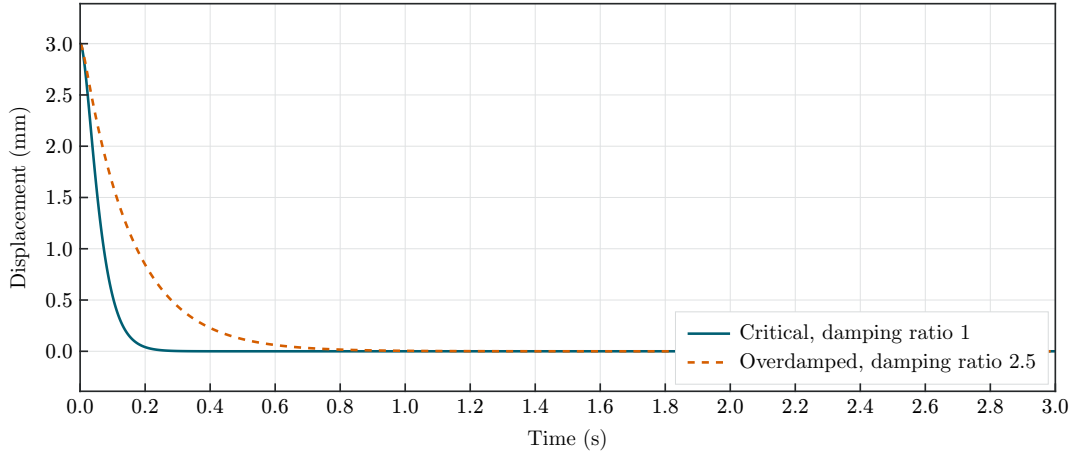


Figure 13: B.5.3.e. Overdamped and critical responses over $0 \leq t \leq 3$ s.

B.5.4. Critically damped response with $\zeta = 1$.

(a) **Partial solutions.** The positive root stated in the prose on course-note PDF page 36 is a sign error; its decaying solutions require a negative root. At $\zeta = 1$, the two roots merge into the repeated root $s = -\omega_n$. One solution is the exponential associated with that root. A second independent solution is obtained by multiplying the exponential by t .

$$\boxed{u_1(t) = e^{-\omega_n t}, \quad u_2(t) = t e^{-\omega_n t}} \quad (\text{B.5.4.a})$$

(b) **Complete solution.** The complete response is the linear combination of the two independent partial solutions.

$$\boxed{u(t) = (C_1 + C_2 t) e^{-\omega_n t}} \quad (\text{B.5.4.b})$$

(c) **Initial-condition constants.** At $t = 0$, the displacement condition gives $C_1 = u_0$. Differentiating the complete solution gives

$$\dot{u}(t) = [C_2 - \omega_n(C_1 + C_2 t)] e^{-\omega_n t}.$$

Evaluating at $t = 0$ gives $\dot{u}_0 = C_2 - \omega_n C_1$. Therefore,

$$\boxed{C_1 = u_0, \quad C_2 = \dot{u}_0 + \omega_n u_0} \quad (\text{B.5.4.c})$$

(d) **Numerical values.** The constants are

$$C_1 = 3.000 \text{ mm}, \quad C_2 = 0 + (31.4159)(3) = 94.2478 \text{ mm/s}.$$

$$\boxed{C_1 = 3.00 \text{ mm}, \quad C_2 = 94.25 \text{ mm/s}} \quad (\text{B.5.4.d})$$

The critically damped displacement is consequently

$$u_{\text{CD}}(t) = (3 + 94.2478t) e^{-31.4159t} \text{ mm}.$$

(e) Critically damped plot. The shorter time view makes the difference between the two nonoscillatory responses visible near the initial displacement.

B5. Critical and overdamped displacement release

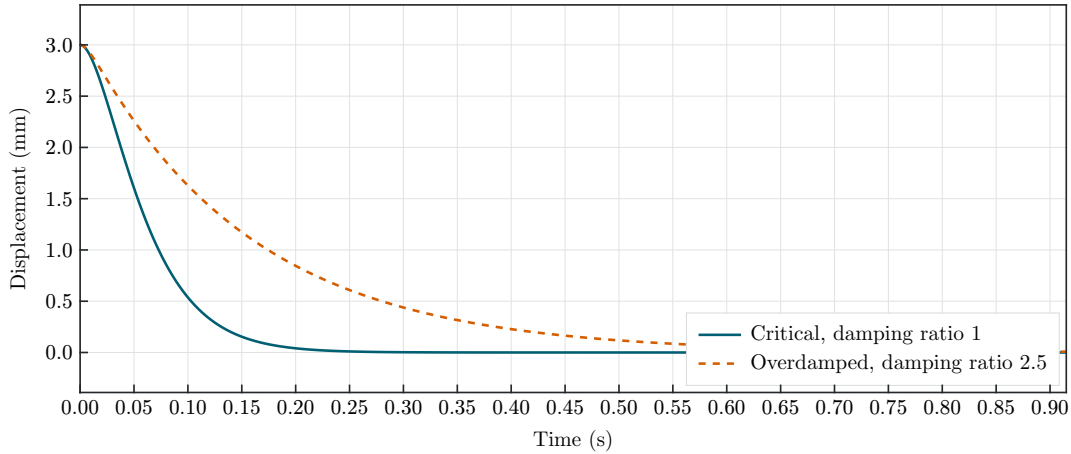


Figure 14: B.5.4.e. Detailed comparison of the critically damped and overdamped responses.

B.5.6. Critical versus overdamped design.

Both designs return to equilibrium without oscillation. Critical damping has the repeated decay rate $-\omega_n$, so it gives the fastest nonoscillatory return for the specified ω_n and initial conditions. Its main disadvantage is that the damping must be tuned close to the critical value. Changes in mass, stiffness, or damper properties move the system away from that condition.

For $\zeta = 2.5$, the overdamped response contains a fast rate of -150.523 s^{-1} and a slow rate of -6.55689 s^{-1} . The fast exponential has a negative coefficient, so that term increases toward zero as it decays. The derivatives of the two terms cancel at release to give the specified zero initial velocity. The qualitative initial-decay statement on course-note PDF pages 38–39 does not apply to this zero-velocity release. For these initial conditions, the critically damped response returns faster both initially and at late times; the slow overdamped root controls the long tail. For example, at $t = 0.010 \text{ s}$, the critical response is 2.880 mm , while the overdamped response is 2.907 mm . Overdamping can therefore provide more tolerance against crossing into oscillatory behavior, but it sacrifices response speed and can increase damper forces during rapid motion. A race-car suspension must balance these effects with tire contact, road inputs, suspension travel, and damper-force limits.

MATLAB code. *MATLAB script for the B.5 overdamped and critically damped response plots.*

```

01 % B.5. Equal initial conditions for critical and overdamped motion.
02 cfg = hw01_settings();
03 fn = 5; u0 = 3/cfg.mmPerM; v0 = 0; % Hz, m, m/s
04 zeta = [1 2.5]; tEnd = 3; % Required duration, s.
05 disp(table(fn,u0,v0,tEnd)); disp(zeta);
06 wn = 2*pi*fn;
07 q = sqrt(zeta(2)^2-1);
08 sFast = -wn*(zeta(2)+q); sSlow = -wn/(zeta(2)+q);
09 dt = 1/(abs(sSlow)*cfg.samplesPerDecayTime);
10 fastEnd = min(tEnd,6/abs(sFast)); % Resolve the first six fast decay times.
11 fastGrid = linspace(0,fastEnd,6*cfg.samplesPerDecayTime+1);
12 t = unique([linspace(0,tEnd,ceil(tEnd/dt)+1),fastGrid]);
13 u = zeros(numel(zeta),numel(t));
14 for j=1:numel(zeta)
15     u(j,:) = hw01_free(t,wn,zeta(j),u0,v0);
16     hw01_check_initial(wn,zeta(j),u0,v0);
17 end
18 disp(table(wn,sFast,sSlow));
19 earlyTime = .01; % Evaluation time used in the written discussion, s.
20 earlyU = [hw01_free(earlyTime,wn,1,u0,v0), ...
21     hw01_free(earlyTime,wn,zeta(2),u0,v0)]*cfg.mmPerM;
22 disp(table(earlyTime,earlyU(1),earlyU(2),'VariableNames', ...
23     {'time_s','critical_mm','overdamped_mm'}));
24 hw01_plot('B5',t,u*cfg.mmPerM,[],[],[]);
25 detailEnd = min(tEnd,6/abs(sSlow)); % Six slow decay times.
26 td = unique([linspace(0,detailEnd,ceil(detailEnd/dt)+1), ...
27     fastGrid(fastGrid<=detailEnd)]);
28 ud = [hw01_free(td,wn,1,u0,v0); ...
29     hw01_free(td,wn,zeta(2),u0,v0)];
30 hw01_plot('B5_detail',td,ud*cfg.mmPerM,[],[],[]);

```

B.7 Calculate Flex-Beam Response to Slightly Unbalanced Electric Motor Excitation

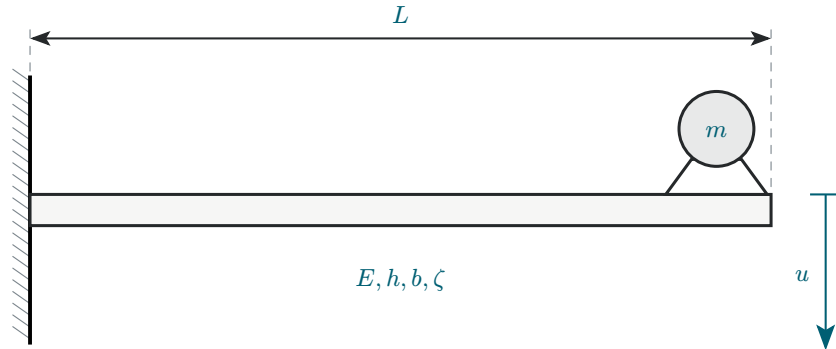


Figure 15: Damped cantilever beam excited by force $F(t)$ from slightly unbalanced electric motor.

Consider a cantilever beam of length $L = 520 \text{ mm}$, elastic modulus $E = 200 \text{ GPa}$, cross-section dimensions $h = 4 \text{ mm}$ and $b = 30 \text{ mm}$, and damping ratio $\zeta = 15\%$. (The damping is achieved through a proprietary damping treatment applied on the beam surface.) An electric motor of mass $m = 0.7 \text{ kg}$ operating at $3,100 \text{ RPM}$ is attached at the beam tip as shown in Figure 15. The electric motor is slightly unbalanced, producing a slight vibratory excitation $F(t) = \hat{F}e^{i\omega t}$ with $\hat{F} = 12 \text{ mN}$. The excitation frequency ω depends on the electric motor speed.

Do the following.

- (1) Display input data.
- (2) Calculate:
 - (a) Cross-section second moment of area I , flexural stiffness EI , and spring constant k .
 - (b) Natural frequency ω_n rad/s, critical damping c_{cr} , physical damping c .
 - (c) Quasi-static deflection u_{qst} corresponding to $\omega_n \rightarrow 0$.
- (3) Calculate the values of:
 - (a) Resonant frequency ω_r and damped frequency ω_d .
 - (b) Peak magnification M_r .
 - (c) Magnification at phase resonance M_{90} .
 - (d) Physical response $\hat{u}_{90}$ at phase resonance and its magnitude $|\hat{u}_{90}|$.
 - (e) Calculate the Hz frequencies f_r , f_n and the corresponding RPM values.
- (4) Assume excitation at phase resonance:
 - (a) Calculate the excitation frequency ω in rad/s and f in Hz.
 - (b) Calculate period of oscillation τ .
 - (c) Plot the time response for $N_{osc} = 13$ oscillations. Hint: plot the real part of the complex displacement $u(t)$.
- (5) Graduate work. Complete items (a)–(d).

- (a) Plot the magnitude and phase of the frequency response function FRF vs. excitation f in Hz from quasi-static to twelve times the phase-resonance frequency.
- (b) Calculate the FRF magnitude at phase-resonance RPM and at double the phase-resonance RPM.
- (c) Calculate the vibration magnitude $|u_0|$ when the electric motor runs at its operational RPM.
- (d) Comment on what vibration would be observed as the motor is started and accelerated to reach its operational RPM.

Solution to B.7

B.7.1. Input data. I model the cantilever by its equivalent tip stiffness and use the attached motor mass as the lumped vibrating mass. I omit beam mass because neither beam density nor beam mass is supplied. Since h is the bending depth for vertical bending, the second moment of area is $I = bh^3/12$.

Quantity	Value	Unit
L	0.520	m
E	200	GPa
h	4.00	mm
b	30.0	mm
m	0.700	kg
ζ	0.150	1
$\hat{F}$	12.0	mN
N_0	3100	rpm
N_{osc}	13	oscillations
N_f	5000	points

B.7.2. Beam and damping parameters.

(a) Section and beam stiffness. Because the beam bends about the weak axis, the rectangular-section second moment of area is

$$I = \frac{bh^3}{12} = \frac{(0.030)(0.004)^3}{12} = 1.600 \times 10^{-10} \text{ m}^4 = 160.0 \text{ mm}^4.$$

The tip stiffness of a fixed-free beam is $k = 3EI/L^3$. Therefore,

$$EI = (200 \times 10^9)(1.600 \times 10^{-10}) = 32.00 \text{ N m}^2, \quad k = \frac{3(32.00)}{(0.520)^3} = 682.75 \text{ N/m}.$$

$$\boxed{I = 1.600 \times 10^{-10} \text{ m}^4, \quad EI = 32.00 \text{ N m}^2, \quad k = 682.75 \text{ N/m}} \quad (\text{B.7.2.a})$$

(b) Natural frequency and damping. The natural frequency follows from the lumped mass and beam stiffness.

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{682.75}{0.700}} = 31.2307 \text{ rad/s}, \quad f_n = \frac{\omega_n}{2\pi} = 4.9705 \text{ Hz}.$$

Critical damping is $c_{\text{cr}} = 2m\omega_n$. Since the damping ratio is $\zeta = c/c_{\text{cr}}$, the physical damping coefficient is

$$c_{\text{cr}} = 2(0.700)(31.2307) = 43.723 \text{ N s/m}, \quad c = \zeta c_{\text{cr}} = (0.150)(43.723) = 6.5584 \text{ N s/m}.$$

$$\begin{aligned} \omega_n &= 31.2307 \text{ rad/s}, & f_n &= 4.9705 \text{ Hz}, \\ c_{\text{cr}} &= 43.723 \text{ N s/m}, & c &= 6.5584 \text{ N s/m} \end{aligned} \quad (\text{B.7.2.b})$$

(c) Quasi-static deflection. Because quasi-static loading makes the excitation frequency negligible, $\omega \rightarrow 0$ while ω_n remains fixed. The force-normalized frequency response then approaches one, so

$$u_{\text{qst}} = \frac{\hat{F}}{k} = \frac{0.012}{682.75} = 1.7576 \times 10^{-5} \text{ m.}$$

$$u_{\text{qst}} = 17.576 \text{ } \mu\text{m} \quad (\text{B.7.2.c})$$

B.7.3. Resonance quantities.

(a) True resonance and damped frequency. Since $\zeta < 1/\sqrt{2}$, the magnification has a nonzero maximum. Differentiating the magnification with respect to the frequency ratio gives

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} = 30.5199 \text{ rad/s.}$$

The damped natural frequency comes from the complementary solution.

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 30.8773 \text{ rad/s.}$$

$$\omega_r = 30.5199 \text{ rad/s}, \quad \omega_d = 30.8773 \text{ rad/s} \quad (\text{B.7.3.a})$$

(b) Peak magnification. Substituting ω_r into the magnification expression gives

$$M_r = \frac{1}{2\zeta\sqrt{1 - \zeta^2}} = \frac{1}{2(0.150)\sqrt{1 - (0.150)^2}} = 3.37148.$$

$$M_r = 3.37148 \quad (\text{B.7.3.b})$$

(c) Magnification at phase resonance. Phase resonance occurs when the real part of the FRF denominator is zero, which gives $\omega = \omega_n$. Thus,

$$M_{90} = \frac{1}{2\zeta} = \frac{1}{2(0.150)} = 3.33333.$$

$$\boxed{M_{90} = 3.33333} \quad (\text{B.7.3.c})$$

(d) **Physical response at phase resonance.** At $\omega = \omega_n$, the normalized FRF is

$$\text{FRF}(\omega_n) = \frac{1}{2i\zeta} = -\frac{i}{2\zeta}.$$

Multiplying by the quasi-static displacement gives

$$\hat{u}_{90} = u_{\text{qst}} \text{FRF}(\omega_n) = -i \frac{u_{\text{qst}}}{2\zeta} = -i 58.5867 \mu\text{m}.$$

Therefore,

$$\boxed{\hat{u}_{90} = -i 58.5867 \mu\text{m}, \quad |\hat{u}_{90}| = 58.5867 \mu\text{m}} \quad (\text{B.7.3.d})$$

(e) **Frequency and RPM conversions.** Since $f = \omega/(2\pi)$, division by 2π converts each angular frequency to hertz. Multiplication by 60 then converts hertz to revolutions per minute.

$$\begin{aligned} f_r &= \frac{\omega_r}{2\pi} = 4.85739 \text{ Hz}, & N_r &= 60f_r = 291.444 \text{ rpm}, \\ f_n &= \frac{\omega_n}{2\pi} = 4.97052 \text{ Hz}, & N_{90} &= 60f_n = 298.231 \text{ rpm}. \end{aligned}$$

$$\boxed{\begin{aligned} f_r &= 4.85739 \text{ Hz}, & N_r &= 291.444 \text{ rpm}, \\ f_n &= 4.97052 \text{ Hz}, & N_{90} &= 298.231 \text{ rpm} \end{aligned}} \quad (\text{B.7.3.e})$$

B.7.4. Time response at phase resonance.

(a) **Excitation frequency.** Phase resonance uses $\omega = \omega_n$, so

$$\boxed{\omega = 31.2307 \text{ rad/s}, \quad f = \frac{\omega}{2\pi} = 4.97052 \text{ Hz}} \quad (\text{B.7.4.a})$$

(b) **Period.** The period of the phase-resonant oscillation is

$$\tau = \frac{2\pi}{\omega} = \frac{2\pi}{31.2307} = 0.201186 \text{ s}.$$

$$\boxed{\tau = 0.201186 \text{ s}, \quad N_{\text{osc}}\tau = 2.61542 \text{ s}} \quad (\text{B.7.4.b})$$

(c) **Real displacement.** The physical force is taken as $F(t) = \hat{F} \cos(\omega_n t)$. Since $\hat{u}_{90} = -iU$, where

$$U = \frac{u_{\text{qst}}}{2\zeta} = 58.5867 \mu\text{m},$$

the real part of the complex steady-state response is

$$u_{\text{ss}}(t) = \text{Re}\{\hat{u}_{90}e^{i\omega_n t}\} = U \sin(\omega_n t).$$

The assignment specifies the number of cycles but not initial conditions for this plot, so the steady-state response is the direct result. For comparison, the figure also shows the response when the force is switched on at $t = 0$ and the mass starts from rest. Applying $u(0) = 0$ and $\dot{u}(0) = 0$ to the complete response gives

$$u_{\text{rest}}(t) = U \sin(\omega_n t) - \frac{U\omega_n}{\omega_d} e^{-\zeta\omega_n t} \sin(\omega_d t).$$

B7. Phase-resonant response

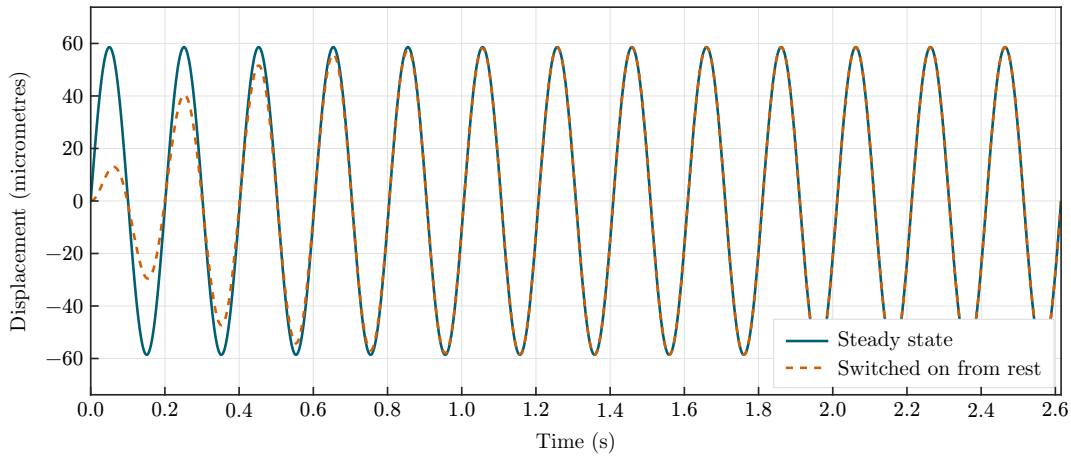


Figure 16: B.7.4.c. Phase-resonant real displacement over $N_{\text{osc}} = 13$ cycles. The dashed orange curve includes the transient required when the force is switched on from rest.

B.7.5. Graduate work.

(a) **Operational frequency and FRF sweep.** The operating speed is $N_0 = 3100$ rpm, so

$$f_0 = \frac{N_0}{60} = 51.6667 \text{ Hz}, \quad \omega_0 = 2\pi f_0 = 324.631 \text{ rad/s}.$$

For the sweep, define $p = \omega/\omega_n = 2\pi f/\omega_n$. The magnitude and phase are

$$M(f) = \frac{1}{\sqrt{(1-p^2)^2 + (2\zeta p)^2}}, \quad \phi(f) = -\text{atan2}(2\zeta p, 1-p^2).$$

Using $N_f = 5000$ uniformly spaced frequencies from $f = 0$ to $f = 12f_n = 59.6462$ Hz gives the magnitude and phase shown below.

B7. Normalized frequency response, 5000 samples

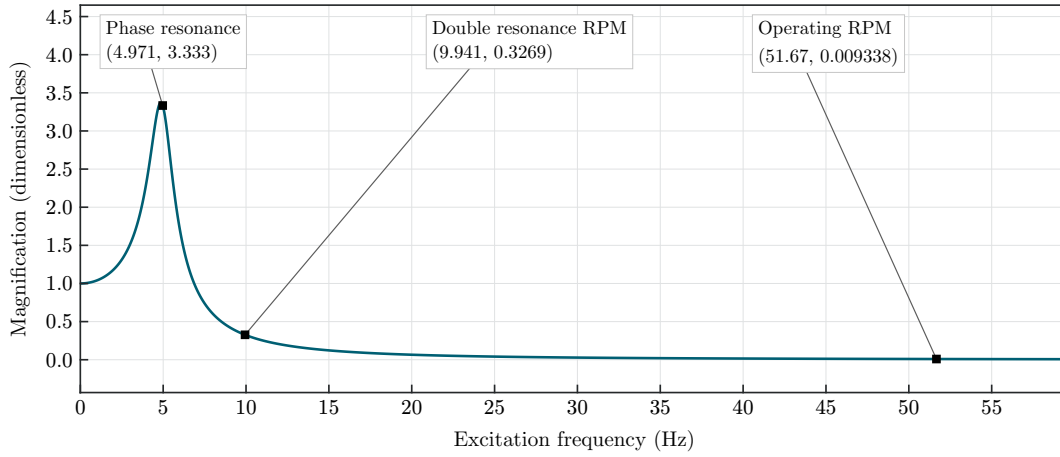


Figure 17: *B.7.5.a. Dimensionless FRF magnitude from 0 to $12f_n = 59.6462$ Hz.*

B7. Response phase relative to force

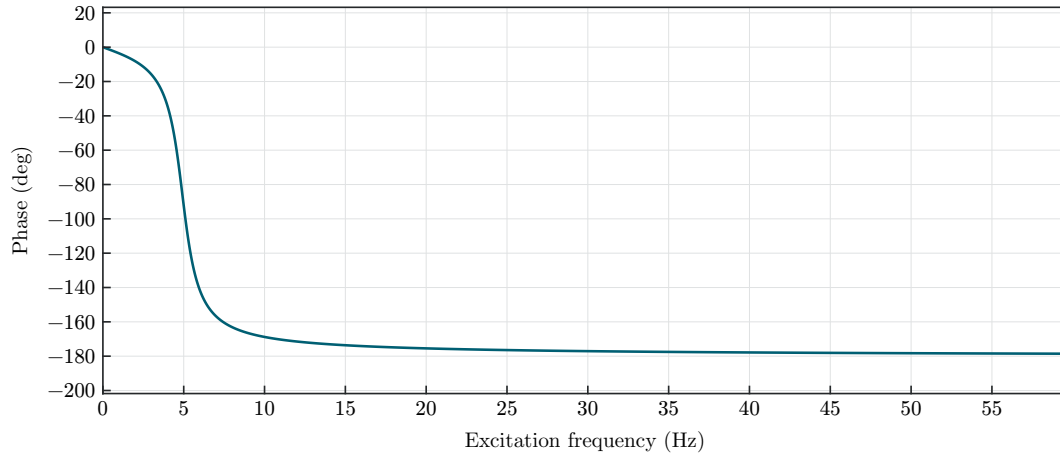


Figure 18: *B.7.5.a. FRF phase from 0 to $12f_n = 59.6462$ Hz.*

(b) Requested FRF values. The phase-resonance speed is $N_{90} = 298.231$ rpm. Doubling that speed doubles the excitation frequency, so the second requested point is $2N_{90} = 596.462$ rpm and $f = 2f_n = 9.94104$ Hz.

Operating point	f (Hz)	N (rpm)	$ FRF $
Phase resonance	4.97052	298.231	3.33333
Double phase-resonance speed	9.94104	596.462	0.32686
Operational speed	51.6667	3100.000	0.009338

(c) Operational vibration magnitude. At $f_0 = 51.6667$ Hz, the frequency ratio is $p_0 = \omega_0/\omega_n = 10.39462$. Therefore,

$$|\hat{u}_0| = u_{\text{qst}} M(f_0) = (17.576 \mu\text{m})(0.00933763) = 0.164118 \mu\text{m}.$$

$$\boxed{|\hat{u}_0| = 0.164118 \mu\text{m}} \quad (\text{B.7.5.c})$$

(d) Run-up interpretation. When the motor starts, the displacement contains a transient because the mass begins away from the eventual forced steady state. As the motor accelerates, the excitation frequency passes through true resonance near $N_r = 291.444 \text{ rpm}$, where the steady-state magnification is largest. Phase resonance occurs slightly later at $N_{90} = 298.231 \text{ rpm}$, so the two speeds are close but not identical.

After the motor passes this region, the FRF magnitude decreases and the displacement becomes small at the operational speed. Under the supplied constant force amplitude, the predicted steady-state magnitude decreases from about $58.6 \mu\text{m}$ near phase resonance to $0.164 \mu\text{m}$ at 3100 rpm . The actual maximum during acceleration also depends on how quickly the motor crosses resonance. A rotating imbalance normally produces a force that varies with speed squared, so the fixed- $\hat{F}$ sweep describes the structural FRF and does not by itself predict the complete transient run-up.

MATLAB code. *Time response and FRF plots for B.7.*

```

01 % B.7. Constant force amplitude, as specified in the assignment.
02 cfg = hw01_settings();
03 L=.520; E=200e9; h=.004; b=.030; % m, Pa, m, m
04 m=.70; zeta=.15; Fhat=.012; % kg, dimensionless, N
05 N0=3100; Nosc=13; Nf=5000; % rpm, cycles, requested samples
06 disp(table(L,E,h,b,m,zeta,Fhat,N0,Nosc,Nf));
07 I=b*h^3/12; k=3*E*I/L^3; % Cantilever tip stiffness, N/m.
08 wn=sqrt(k/m); wd=wn*sqrt(1-zeta^2);
09 cCritical=2*m*wn; c=zeta*cCritical;
10 uqst=Fhat/k; U=uqst/(2*zeta); tau=2*pi/wn;
11 wr=wn*sqrt(1-2*zeta^2);
12 Mr=1/(2*zeta*sqrt(1-zeta^2));
13 disp(table(I,k,wn,wd,cCritical,c,uqst,wr,Mr,U,tau));
14 t=hw01_grid(Nosc*tau,tau,[]);
15 uss=U*sin(wn*t);
16 % Cancel the steady-state initial velocity to obtain start from rest.
17 transient=hw01_free(t,wn,zeta,0,-U*wn);
18 urest=uss+transient;
19 [uc0,vc0]=hw01_free(0,wn,zeta,0,-U*wn);
20 assert(abs(uc0)<cfg.checkTolerance);
21 assert(abs(U*wn+vc0)<cfg.checkTolerance);
22 hw01_plot('B7_time',t,[uss;urest]*cfg.umPerM,[],[],[]);
23 f0=N0/cfg.secondsPerMinute; fn=wn/(2*pi);
24 f=linspace(0,12*fn,Nf); p=2*pi*f/wn;
25 FRF=@(r) 1./(1-r.^2+2i*zeta*r);
26 H=FRF(p);
27 eventF=[fn 2*fn f0]; eventH=FRF(eventF/fn);
28 disp(table(eventF',eventF'*cfg.secondsPerMinute, ...
29     abs(eventH)',rad2deg(angle(eventH))', ...
30     uqst*abs(eventH)*cfg.umPerM,'VariableNames', ...
31     {'frequency_Hz','speed_rpm','magnification','phase_deg','amplitude_um'}));
32 hw01_plot('B7_magnitude',f,abs(H),eventF,abs(eventH),[]);
33 hw01_plot('B7_phase',f,rad2deg(angle(H)),[],[],[]);

```

B.8 Graduate Work. Calculate Horizontal Flex-Beam Response to Impact

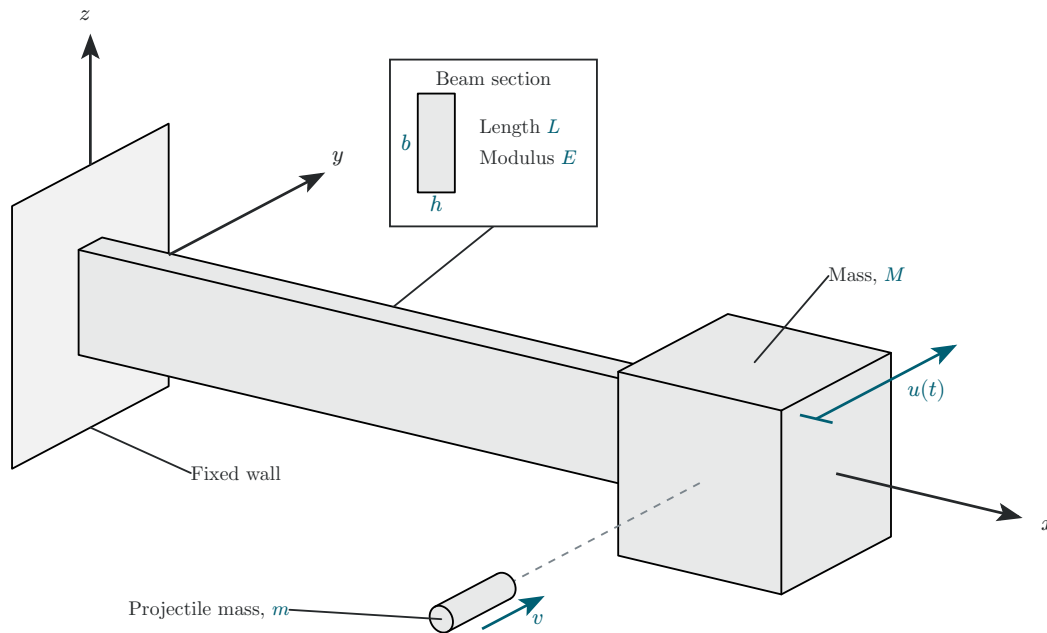


Figure 19: Impact excitation of a horizontal flex-beam vibration system.

Consider a cantilever beam of length $L = 525 \text{ mm}$ with a mass $M = 62 \text{ kg}$ attached at its tip as shown in Figure 19. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 4 \text{ mm}$, $b = 30 \text{ mm}$. The beam is initially at rest. A small projectile of mass $m = 13 \text{ g}$ and velocity $v = 1020 \text{ m/s}$ hits the mass and remains embedded in it. This impact imparts an initial velocity to the system that starts to oscillate with motion $u(t)$ contained in the horizontal plane.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.
- (3) Calculate the cross-sectional stiffness EI of the beam.
- (4) Calculate the flex-beam spring constant k .
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (6) Calculate the rad/sec natural frequency ω_n and period of oscillation τ .
- (7) Calculate the position of the mass $u(t_1)$ after $t_1 = 17 \text{ s}$.
- (8) Plot the response of the system for $N_{\text{osc}} = 13$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response calculated at item (7).

- (9) Discuss your results.

Solution to B.8

B.8.1. Input data. The projectile mass is denoted by m_p to distinguish it from the attached mass M . Since the projectile remains embedded after impact, the vibrating mass is $m_T = M + m_p$. I represent the beam by its equivalent tip stiffness.

Quantity	Value	Unit
L	0.525	m
E	200	GPa
h	4.00	mm
b	30.0	mm
M	62.0	kg
m_p	0.0130	kg
v_p	1020	m/s
t_1	17.0	s
N_{osc}	13	oscillations

B.8.2. Second moment of area. The beam moves horizontally, but the bending axis is determined by the cross-section orientation. In the section shown, h is the bending depth and b is the dimension normal to that depth. Therefore,

$$I = \frac{bh^3}{12} = \frac{(0.030)(0.004)^3}{12} = 1.600 \times 10^{-10} \text{ m}^4 = 160.0 \text{ mm}^4.$$

$$\boxed{I = 1.600 \times 10^{-10} \text{ m}^4 \quad (160.0 \text{ mm}^4)} \quad (\text{B.8.2})$$

B.8.3. Cross-sectional stiffness. Multiplying the elastic modulus by the second moment of area gives

$$EI = (200 \times 10^9)(1.600 \times 10^{-10}) = 32.00 \text{ N m}^2.$$

$$\boxed{EI = 32.00 \text{ N m}^2} \quad (\text{B.8.3})$$

B.8.4. Flex-beam spring constant. The fixed-free beam tip deflection under a concentrated force gives the equivalent spring constant

$$k = \frac{3EI}{L^3} = \frac{3(32.00)}{(0.525)^3} = 663.427 \text{ N/m}.$$

$$\boxed{k = 663.427 \text{ N/m}} \quad (\text{B.8.4})$$

B.8.5. Initial velocity after impact. The collision duration is short compared with the vibration period, so the beam spring force contributes negligible impulse during the collision. Conservation of linear momentum therefore gives

$$m_p v_p + M(0) = (M + m_p) v_0.$$

Solving for the common velocity immediately after impact gives

$$v_0 = \frac{m_p v_p}{M + m_p} = \frac{(0.0130)(1020)}{62.013} = 0.213826 \text{ m/s}.$$

$$\boxed{\dot{u}_0 = v_0 = 0.213826 \text{ m/s}} \quad (\text{B.8.5})$$

B.8.6. Natural frequency and period. After the collision, the projectile and attached mass move together. Because the beam is horizontal and damping is neglected, the free-vibration equation is

$$(M + m_p)\ddot{u} + ku = 0.$$

The natural frequency follows from the characteristic equation.

$$\omega_n = \sqrt{\frac{k}{M + m_p}} = \sqrt{\frac{663.427}{62.013}} = 3.27081 \text{ rad/s}.$$

The period is the time required for one complete cycle.

$$\tau = \frac{2\pi}{\omega_n} = \frac{2\pi}{3.27081} = 1.92099 \text{ s}.$$

$$\boxed{\omega_n = 3.27081 \text{ rad/s}, \quad \tau = 1.92099 \text{ s}} \quad (\text{B.8.6})$$

B.8.7. Displacement at $t_1 = 17 \text{ s}$. The beam displacement is continuous through the impact, so $u(0) = 0$. The initial velocity is $\dot{u}(0) = v_0$. Applying these conditions to the undamped free-vibration solution gives

$$u(t) = u(0) \cos(\omega_n t) + \frac{\dot{u}(0)}{\omega_n} \sin(\omega_n t) = \frac{v_0}{\omega_n} \sin(\omega_n t).$$

At $t_1 = 17 \text{ s}$,

$$u(t_1) = \frac{0.213826}{3.27081} \sin((3.27081)(17)) = -0.0529817 \text{ m.}$$

$$\boxed{u(17 \text{ s}) = -52.9817 \text{ mm}} \quad (\text{B.8.7})$$

B.8.8. Thirteen-oscillation response.

(a) **Largest response.** The response amplitude is

$$|u|_{\max} = \frac{v_0}{\omega_n} = \frac{0.213826}{3.27081} = 0.0653741 \text{ m.}$$

The first positive maximum occurs when $\omega_n t = \pi/2$, so

$$t_{\text{peak},1} = \frac{\pi}{2\omega_n} = 0.480247 \text{ s.}$$

$$\boxed{|u|_{\max} = 65.374 \text{ mm} \quad \text{at} \quad t_{\text{peak},1} = 0.480247 \text{ s}} \quad (\text{B.8.8.a})$$

(b) **Plot markers.** Thirteen oscillations span

$$t_{\text{end}} = N_{\text{osc}}\tau = 13(1.92099) = 24.9728 \text{ s.}$$

The plot marks the first positive maximum and the response at $t_1 = 17 \text{ s}$. Their coordinates are (0.480 s, 65.4 mm) and (17.00 s, -53.0 mm).

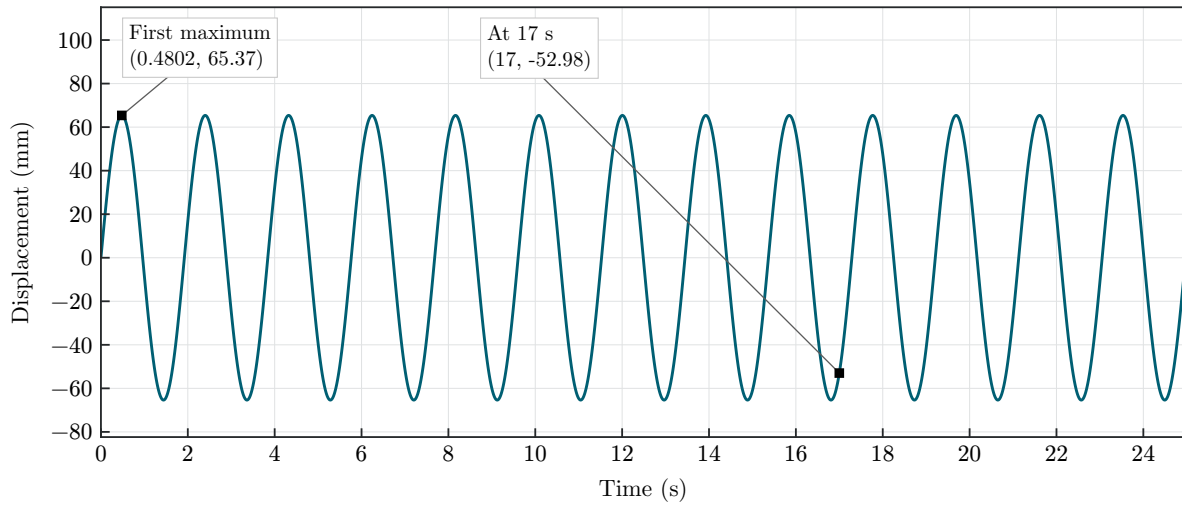
B8. Response after an embedded-projectile impact

Figure 20: *B.8.8.a-b. Undamped horizontal flex-beam response over $N_{\text{osc}} = 13$ cycles.*

B.8.9. Discussion. The projectile transfers its momentum to the combined mass, which creates the initial velocity v_0 while leaving the displacement equal to zero. The undamped model then preserves the amplitude at approximately 65.4 mm. This is about 12.5% of the 525 mm span, so the result is large enough that small-deflection beam theory should be treated as a linear-model estimate.

The negative value at 17 s is caused by the phase of the sinusoidal response at that time; it does not indicate a change in amplitude. The calculation also omits distributed beam mass, damping, local impact deformation, and large-deflection geometric effects. Including those effects would change the natural frequency and reduce or otherwise modify the observed response.

MATLAB code. *Thirteen-oscillation response after the embedded-projectile impact.*

```

01 % B.8. Horizontal cantilever response after an embedded impact.
02 cfg=hw01_settings();
03 L=.525; E=200e9; h=.004; b=.030; % m, Pa, m, m
04 M=62; mp=.013; vp=1020; % kg, kg, m/s
05 t1=17; Nosc=13; u0=0; % s, cycles, m
06 disp(table(L,E,h,b,M,mp,vp,t1,Nosc,u0));
07 I=b*h^3/12; k=3*E*I/L^3;
08 totalMass=M+mp; v0=mp*vp/totalMass;
09 wn=sqrt(k/totalMass); tau=2*pi/wn;
10 amplitude=v0/wn; eventT=[tau/4 t1];
11 t=hw01_grid(Nosc*tau,tau,eventT);
12 u=hw01_free(t,wn,0,u0,v0);
13 eventU=hw01_free(eventT,wn,0,u0,v0);
14 hw01_check_initial(wn,0,u0,v0);
15 assert(abs(totalMass*v0-mp*vp)<cfg.checkTolerance);
16 disp(table(I,k,v0,wn,tau,amplitude*cfg.mmPerM, ...
17     eventU(2)*cfg.mmPerM,'VariableNames', ...
18     {'I_m4','k_N_m','v0_m_s','wn_rad_s','period_s','amplitude_mm','u17_mm'}));
19 hw01_plot('B8',t,u*cfg.mmPerM,eventT,eventU*cfg.mmPerM,[]);

```

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.4 Calculate Horizontal Flex-Beam Response to Initial Displacement

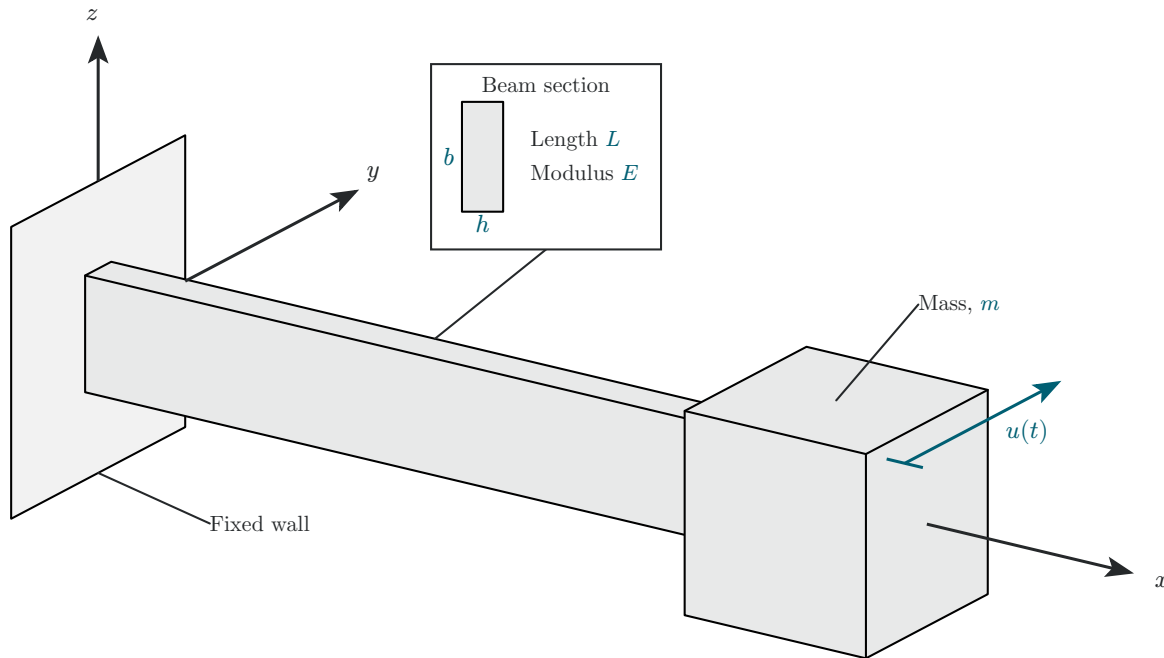


Figure 21: Flex-beam vibrating system.

Consider a cantilever beam of length $L = 525 \text{ mm}$ with a mass $m = 62 \text{ kg}$ attached at its tip as shown in Figure 21. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 4 \text{ mm}$, $b = 30 \text{ mm}$. The tip mass is given an initial displacement $u_0 = 3 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.
- (3) Calculate the cross-sectional stiffness EI of the beam.
- (4) Calculate the spring constant k of the system.
- (5) Calculate the natural frequency in rad/sec ω_n .
- (6) Calculate the natural frequency in Hz f_n .
- (7) Calculate the period of oscillation in sec τ .
- (8) Calculate the position of the mass $u(t_1)$ after $t_1 = 18 \text{ s}$.
- (9) Plot the response of the system for $N_{\text{osc}} = 13$ oscillations and use a data tip to find $u(t_1)$ at

t_1 .

(10) Discuss your results.

Solution to C.4

Using SI units, the given system data are

$$L = 0.525 \text{ m}, \quad m = 62 \text{ kg}, \quad E = 200 \times 10^9 \text{ Pa},$$

$$b = 0.030 \text{ m}, \quad h = 0.004 \text{ m}, \quad u_0 = 0.0030 \text{ m}, \quad v_0 = 0.$$

These values use SI units throughout, so the displacement response will first be calculated in metres.

$$\boxed{\begin{array}{l} L = 0.525 \text{ m}, \quad m = 62 \text{ kg}, \quad E = 200 \text{ GPa}, \\ b = 0.030 \text{ m}, \quad h = 0.004 \text{ m}, \quad u_0 = 3.0 \text{ mm}, \quad v_0 = 0 \end{array}} \quad (\text{C.4.1})$$

For the rectangular flex-beam cross-section, the second moment of area about the bending axis is

$$I = \frac{bh^3}{12} = \frac{(0.030)(0.004)^3}{12} = 1.600 \times 10^{-10} \text{ m}^4.$$

Because $1 \text{ m}^4 = 10^{12} \text{ mm}^4$, this is 160.0 mm^4 .

$$\boxed{I = 1.600 \times 10^{-10} \text{ m}^4 = 160.0 \text{ mm}^4} \quad (\text{C.4.2})$$

The flexural rigidity follows by multiplying the second moment of area by Young's modulus. Therefore,

$$EI = (200 \times 10^9)(1.600 \times 10^{-10}) = 32.00 \text{ N m}^2.$$

$$\boxed{EI = 32.00 \text{ N m}^2} \quad (\text{C.4.3})$$

A cantilever beam with a concentrated tip load has the end displacement $\Delta = PL^3/(3EI)$. Hence its equivalent transverse stiffness is

$$k = \frac{P}{\Delta} = \frac{3EI}{L^3} = \frac{3(32.00)}{(0.525)^3} = 663.4273 \text{ N/m}.$$

$$\boxed{k = 663.43 \text{ N/m}} \quad (\text{C.4.4})$$

With displacement measured from the undeformed reference used for this horizontal system, the undamped equation of motion is $m\ddot{u} + ku = 0$. Dividing by m gives $\ddot{u} + \omega_n^2 u = 0$, where $\omega_n^2 = k/m$. Thus,

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{663.4273}{62}} = 3.27115 \text{ rad/s}.$$

$$\boxed{\omega_n = 3.271 \text{ rad/s}} \quad (\text{C.4.5})$$

The cyclic natural frequency is the angular natural frequency divided by 2π , so

$$f_n = \frac{\omega_n}{2\pi} = \frac{3.27115}{2\pi} = 0.520620 \text{ Hz.}$$

$$\boxed{f_n = 0.5206 \text{ Hz}} \quad (\text{C.4.6})$$

The period is the time for one complete oscillation. Therefore,

$$\tau = \frac{1}{f_n} = \frac{2\pi}{\omega_n} = 1.92079 \text{ s.}$$

$$\boxed{\tau = 1.921 \text{ s}} \quad (\text{C.4.7})$$

The general solution of the undamped equation is

$$u(t) = A \cos(\omega_n t) + B \sin(\omega_n t).$$

At $t = 0$, the initial displacement gives $A = u_0$, while the initial velocity gives $B = v_0/\omega_n$. Since $v_0 = 0$, the response reduces to

$$u(t) = u_0 \cos(\omega_n t) = 0.0030 \cos(3.27115t) \text{ m.}$$

Evaluating this expression at $t_1 = 18 \text{ s}$ gives

$$u(18) = 0.0030 \cos(3.27115(18)) = -0.00206956 \text{ m} = -2.06956 \text{ mm.}$$

$$\boxed{u(18 \text{ s}) = -2.070 \text{ mm}} \quad (\text{C.4.8})$$

For $N_{\text{osc}} = 13$, the plotting interval ends at

$$t_{\text{end}} = N_{\text{osc}}\tau = 13(1.92079) = 24.9702 \text{ s.}$$

The plotted response is shown below. The square marker identifies the calculated value at $t_1 = 18 \text{ s}$, and the same value can be read with a data tip.

C4. Horizontal cantilever displacement release

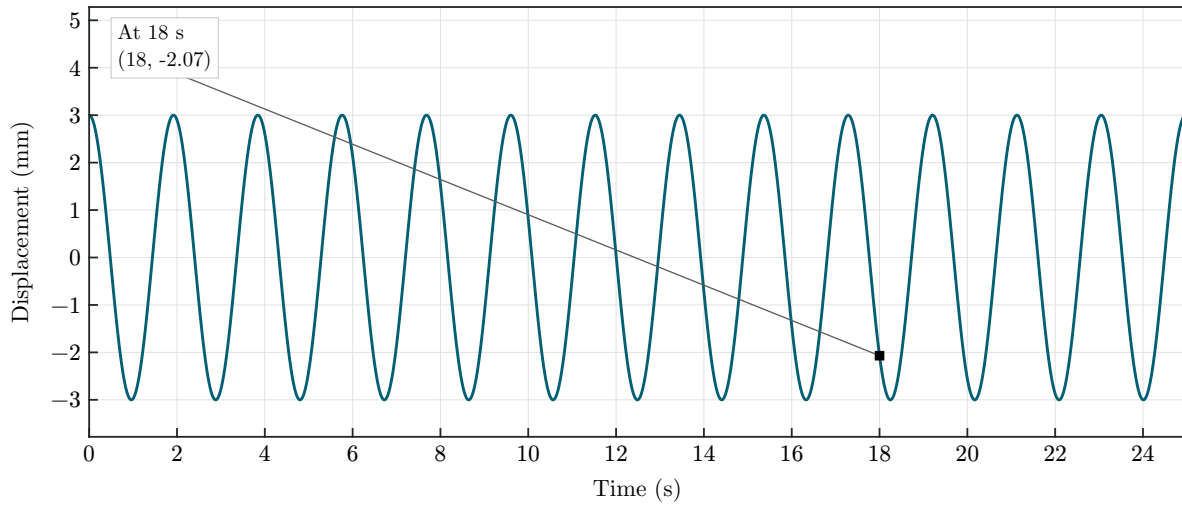


Figure 22: C.4. Thirteen oscillations with the eighteen-second response marked.

MATLAB code. C.4 response plot.

```

01 % C.4. Undamped horizontal cantilever displacement release.
02 cfg=hw01_settings();
03 L=.525; E=200e9; h=.004; b=.030; m=62; % SI units.
04 u0=3/cfg.mmPerM; v0=0; t1=18; Nosc=13;
05 disp(table(L,E,h,b,m,u0,v0,t1,Nosc));
06 I=b*h^3/12; EI=E*I; k=3*EI/L^3;
07 wn=sqrt(k/m); fn=wn/(2*pi); tau=2*pi/wn;
08 tEnd=Nosc*tau;
09 t=hw01_grid(tEnd,tau,t1);
10 u=hw01_free(t,wn,0,u0,v0);
11 u1=hw01_free(t1,wn,0,u0,v0);
12 hw01_check_initial(wn,0,u0,v0);
13 disp(table(I,EI,k,wn,fn,tau,tEnd,u1*cfg.mmPerM, ...
14     'VariableNames',{'I_m4','EI_Nm2','k_N_m','wn_rad_s', ...
15     'fn_Hz','period_s','duration_s','u18_mm'}));
16 hw01_plot('C4',t,u*cfg.mmPerM,t1,u1*cfg.mmPerM,[]);

```

$$\begin{aligned}
 t_{\text{end}} &= 24.9702 \text{ s} \quad \text{for } N_{\text{osc}} = 13, \\
 u_{\text{tip}}(18 \text{ s}) &= -2.070 \text{ mm}
 \end{aligned}
 \tag{C.4.9}$$

The computed displacement and the data-tip value agree because both use the same undamped response. The displacement changes sign as the mass passes through equilibrium, but its maximum absolute value remains $u_0 = 3.0 \text{ mm}$. Since no damping or external force is included, the oscillation continues with constant amplitude and period.

$$\max |u(t)| = 3.0 \text{ mm}, \quad \tau = 1.921 \text{ s with constant amplitude} \tag{C.4.10}$$

C.6 Calculate Cantilever Response to Suddenly Applied Weight

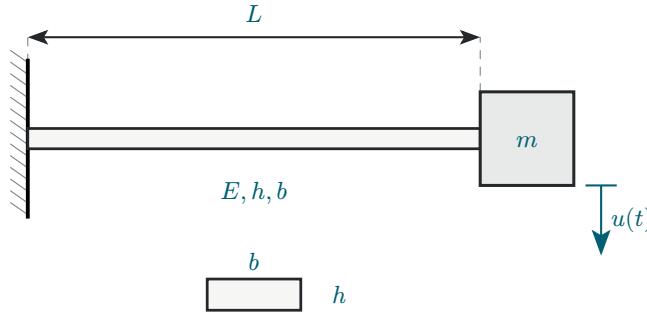


Figure 23: Cantilever beam with tip mass.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a cantilever beam of length $L = 525 \text{ mm}$ and a mass $m = 3 \text{ kg}$ that has to be attached to its tip as shown in Figure 23. The beam is made of steel with elastic modulus $E = 200 \text{ GPa}$, yield stress $Y = 710 \text{ MPa}$, and cross-section dimensions $h = 4 \text{ mm}$ and $b = 30 \text{ mm}$.

Do the following.

- (1) Display input data.
- (2) Calculate the cross-section second moment of area I , the flexural stiffness EI , the spring constant k , the natural frequency ω_n rad/s, f_n Hz, and period of oscillation τ of the system.
- (3) Calculate the weight F_0 of the mass m .
- (4) Static case. Assume the mass is attached to the undeformed beam and then released gradually without inducing vibration. Calculate the static displacement u_{st} .
- (5) Dynamic case. Assume the mass is attached to the undeformed beam and then released suddenly, inducing vibration.
 - (a) Calculate the maximum dynamic displacement u_{dyn} .
 - (b) Plot the response $u(t)$ measured from the undeformed position for $N_{\text{osc}} = 13$ oscillations.
 - (c) Place a data tip on the plot to identify the maximum displacement and compare it with the result of item (5)(a).
- (6) Graduate work. Calculate the maximum stress at the root of the beam and the safety factor $\text{SF} = Y/\sigma$, and comment on your results for the following two cases.
 - (a) Static-case stress σ_{st} .
 - (b) Dynamic-case stress σ_{dyn} .
- (7) Discuss your results.

Solution to C.6

Using SI units, the given parameters are

$$L = 0.525 \text{ m}, \quad m = 3.0 \text{ kg}, \quad E = 200 \times 10^9 \text{ Pa}, \quad h = 0.004 \text{ m}, \quad b = 0.030 \text{ m},$$

$$g = 9.81 \text{ m/s}^2, \quad Y = 710 \times 10^6 \text{ Pa}.$$

The mass is attached at the undeformed tip and is initially at rest.

$$\boxed{\begin{array}{l} L = 0.525 \text{ m}, \quad m = 3.0 \text{ kg}, \quad E = 200 \text{ GPa}, \\ h = 4 \text{ mm}, \quad b = 30 \text{ mm}, \quad g = 9.81 \text{ m/s}^2, \quad Y = 710 \text{ MPa} \end{array}} \quad (\text{C.6.1})$$

The beam bends vertically, so the cross-section dimension h lies in the bending direction. For the rectangular cross-section, the second moment of area is

$$I = \frac{bh^3}{12} = \frac{(0.030)(0.004)^3}{12} = 1.600 \times 10^{-10} \text{ m}^4 = 160.0 \text{ mm}^4.$$

The flexural rigidity is then

$$EI = (200 \times 10^9)(1.600 \times 10^{-10}) = 32.00 \text{ N m}^2.$$

For a cantilever with a concentrated tip load, the end displacement is $\Delta = PL^3/(3EI)$. Therefore, the equivalent spring constant is

$$k = \frac{P}{\Delta} = \frac{3EI}{L^3} = \frac{3(32.00)}{(0.525)^3} = 663.4273 \text{ N/m}.$$

The natural frequency and period follow from

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{663.4273}{3.0}} = 14.8709 \text{ rad/s},$$

$$f_n = \frac{\omega_n}{2\pi} = 2.36677 \text{ Hz}, \quad \tau = \frac{2\pi}{\omega_n} = 0.422517 \text{ s}.$$

$$\boxed{\begin{array}{l} I = 160.0 \text{ mm}^4, \quad EI = 32.00 \text{ N m}^2, \quad k = 663.43 \text{ N/m}, \\ \omega_n = 14.87 \text{ rad/s}, \quad f_n = 2.367 \text{ Hz}, \quad \tau = 0.4225 \text{ s} \end{array}} \quad (\text{C.6.2})$$

The applied weight is

$$F_0 = mg = (3.0)(9.81) = 29.4300 \text{ N}.$$

$$\boxed{F_0 = 29.43 \text{ N}} \quad (\text{C.6.3})$$

For the gradual-release case, the mass reaches equilibrium without oscillating. Setting the acceleration to zero gives

$$ku_{\text{st}} = F_0, \quad u_{\text{st}} = \frac{F_0}{k} = \frac{29.4300}{663.4273} = 0.0443606 \text{ m.}$$

Thus, the static tip displacement is 44.3606 mm.

$$\boxed{u_{\text{st}} = 44.36 \text{ mm}} \quad (\text{C.6.4})$$

For the sudden-release case, the load is applied at $t = 0$ while the mass is still at the undeformed position. The equation of motion for $t > 0$ is

$$m\ddot{u} + ku = F_0.$$

The at-rest initial conditions are

$$u(0) = 0, \quad \dot{u}(0) = 0.$$

Using the constant particular solution $u_p = F_0/k$ and the complementary solution $u_c = A \cos(\omega_n t) + B \sin(\omega_n t)$, these initial conditions give $A = -F_0/k$ and $B = 0$. Therefore, the displacement measured from the undeformed position is

$$u(t) = \frac{F_0}{k} [1 - \cos(\omega_n t)] = 0.0443606 [1 - \cos(14.8709t)] \text{ m.}$$

The largest value occurs when $\cos(\omega_n t) = -1$, which first occurs at

$$t_{\text{max}} = \frac{\pi}{\omega_n} = 0.211258 \text{ s.}$$

Consequently,

$$u_{\text{dyn}} = u_{\text{max}} = \frac{2F_0}{k} = 2u_{\text{st}} = 0.0887211 \text{ m.}$$

In millimetres, this is 88.7211 mm.

$$\boxed{u_{\text{dyn}} = u_{\text{max}} = 88.72 \text{ mm} \quad \text{at} \quad t_{\text{max}} = 0.2113 \text{ s}} \quad (\text{C.6.5.a})$$

For $N_{\text{osc}} = 13$, the plotting interval is

$$t_{\text{end}} = N_{\text{osc}}\tau = 13(0.422517) = 5.49272 \text{ s.}$$

The response plot below is measured from the undeformed beam position. The dashed line is the static displacement, and the square marker identifies the first dynamic maximum.

C6. Ideal linear response to suddenly applied weight

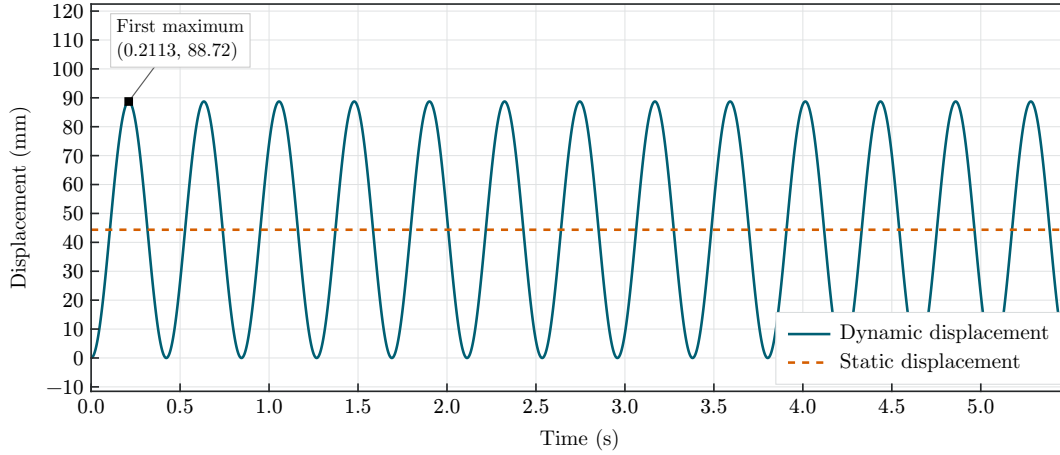


Figure 24: C.6. Thirteen-cycle response measured from the undeformed position.

$$N_{\text{osc}} = 13, \quad t_{\text{end}} = 5.49272 \text{ s} \quad (\text{C.6.5.b})$$

The data tip at the first maximum reads approximately

$$(t_{\text{max}}, u_{\text{max}}) = (0.211 \text{ s}, 88.7 \text{ mm}).$$

This agrees with the analytical value in item (5)(a) to the precision shown.

$$u_{\text{tip}}(0.211 \text{ s}) = 88.7 \text{ mm} \approx u_{\text{dyn}} \quad (\text{C.6.5.c})$$

For the static case, the root bending moment is the applied weight multiplied by the beam length,

$$M_{\text{st}} = F_0 L = (29.4300)(0.525) = 15.45075 \text{ N m}.$$

The outer-fibre distance is $c = h/2 = 0.0020 \text{ m}$. Using the flexure relation $\sigma = Mc/I$, the static root stress is

$$\sigma_{\text{st}} = \frac{M_{\text{st}} c}{I} = \frac{(15.45075)(0.0020)}{1.600 \times 10^{-10}} = 1.93134 \times 10^8 \text{ Pa} = 193.134 \text{ MPa}.$$

The corresponding safety factor is

$$\text{SF}_{\text{st}} = \frac{Y}{\sigma_{\text{st}}} = \frac{710}{193.134} = 3.676.$$

$$\sigma_{\text{st}} = 193.134 \text{ MPa}, \quad \text{SF}_{\text{st}} = 3.676 \quad (\text{C.6.6.a})$$

At the dynamic maximum, the beam restoring force is

$$P_{\text{dyn}} = ku_{\text{dyn}} = 2F_0.$$

Therefore, the dynamic root moment is twice the static root moment,

$$M_{\text{dyn}} = P_{\text{dyn}}L = 2F_0L = 30.9015 \text{ N m}.$$

The maximum dynamic stress and its safety factor are

$$\sigma_{\text{dyn}} = \frac{M_{\text{dyn}}c}{I} = 3.86269 \times 10^8 \text{ Pa} = 386.269 \text{ MPa},$$

$$\text{SF}_{\text{dyn}} = \frac{Y}{\sigma_{\text{dyn}}} = \frac{710}{386.269} = 1.838.$$

$$\boxed{\sigma_{\text{dyn}} = 386.269 \text{ MPa}, \quad \text{SF}_{\text{dyn}} = 1.838} \quad (\text{C.6.6.b})$$

Both idealized safety factors exceed one, so the calculated bending stress remains below the specified yield stress in the static and dynamic cases. Sudden release doubles the tip displacement, root moment, and bending stress and halves the static safety factor. The predicted dynamic displacement is $0.169L$, so geometric nonlinearity and the actual attachment conditions should be checked before treating the linear estimate as a design limit.

$$\boxed{\text{SF}_{\text{dyn}} = 1.838 > 1 \Rightarrow \text{yielding is not predicted by the ideal model}} \quad (\text{C.6.7})$$

MATLAB code. *C.6 response plot.*

```

01 % C.6. Suddenly applied weight and ideal linear bending stresses.
02 cfg=hw01_settings();
03 L=.525; E=200e9; h=.004; b=.030; % m, Pa, m, m
04 m=3; g=9.81; yieldStress=710e6; Nosc=13; % kg, m/s^2, Pa.
05 disp(table(L,E,h,b,m,g,yieldStress,Nosc));
06 I=b*h^3/12; k=3*E*I/L^3; wn=sqrt(k/m);
07 F0=m*g; ust=F0/k; tau=2*pi/wn;
08 tPeak=pi/wn; uPeak=2*ust;
09 t=hw01_grid(Nosc*tau,tau,tPeak);
10 % Shift from the unloaded coordinate to equilibrium before solving.
11 u=ust+hw01_free(t,wn,0,-ust,0);
12 [initialRelative,initialVelocity]=hw01_free(0,wn,0,-ust,0);
13 assert(abs(ust+initialRelative)<cfg.checkTolerance);
14 assert(abs(initialVelocity)<cfg.checkTolerance);
15 outerFiber=h/2;
16 momentStatic=F0*L; momentDynamic=k*uPeak*L;
17 stressStatic=momentStatic*outerFiber/I;
18 stressDynamic=momentDynamic*outerFiber/I;
19 SFstatic=yieldStress/stressStatic;
20 SFdynamic=yieldStress/stressDynamic;
21 disp(table(I,k,wn,tau,ust*cfg.mmPerM,uPeak*cfg.mmPerM, ...
22     stressStatic/1e6,stressDynamic/1e6,SFstatic,SFdynamic, ...
23     'VariableNames',{'I_m4','k_N_m','wn_rad_s','period_s', ...
24     'static_mm','dynamic_mm','static_MPa','dynamic_MPa', ...
25     'SFstatic','SFdynamic'}));
26 fprintf('Dynamic displacement/span = %.4g\n',uPeak/L);
27 fprintf('Linear dynamic stress exceeds yield = %d\n',SFdynamic<1);
28 hw01_plot('C6',t,[u;ust*ones(size(t))]*cfg.mmPerM, ...
29     tPeak,uPeak*cfg.mmPerM,[]);

```

C.8 Calculate Flex-Beam Pendulum Response to Impact

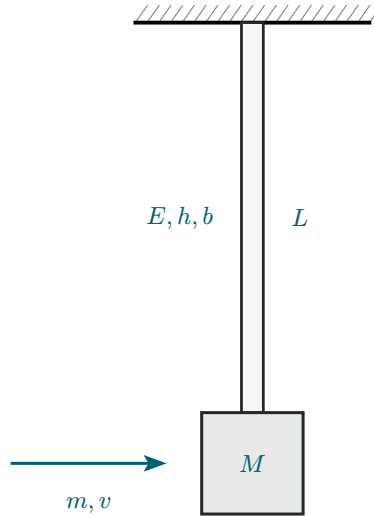


Figure 25: Impact excitation of a vertical flex-beam pendulum.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a vertical flex-beam pendulum of length $L = 440 \text{ mm}$ with a mass $M = 62 \text{ kg}$ attached at its end, as shown in Figure 25. The beam is steel with $E = 200 \text{ GPa}$ and cross-section dimensions $h = 4 \text{ mm}$ and $b = 30 \text{ mm}$. The beam is initially at rest. A projectile of mass $m = 11 \text{ g}$ and velocity $v = 1020 \text{ m/s}$ hits the mass and remains embedded in it. The projectile orientation causes the beam to flex about its minor cross-sectional axis. The impact imparts an initial velocity to the system, which then oscillates.

Do the following.

- (1) Display input data.
- (2) Calculate the minor second moment of area I of the beam.
- (3) Calculate the minor cross-sectional stiffness EI .
- (4) Calculate the flex-beam spring constant k .
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as resulting from the interaction between the projectile and the large mass.
- (6) Calculate the rad/s natural frequency $\omega_{n,\text{FBP}}$ and period of oscillation τ_{FBP} of the flex-beam pendulum with projectile embedded into it.
- (7) Calculate the response $u(t_1)$ after $t_1 = 18 \text{ s}$.
- (8) Plot the response of the system for $N_{\text{osc}} = 23$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response value calculated at item (7).
- (9) Discuss your results in comparison with those of Problem A.3.

Solution to C.8

Denote the projectile mass by m_p and the mass that oscillates after impact by $M_t = M + m_p$. The beam flexes about its minor axis, and the transverse displacement is measured from the vertical position. All calculations use SI units.

Quantity	Value	Unit
L	0.440	m
M	62	kg
m_p	0.011	kg
v_p	1020	m/s
E	200×10^9	Pa
h	0.004	m
b	0.030	m
g	9.81	m/s ²
t_1	18	s
N_{osc}	23	cycles

$$\boxed{M_t = M + m_p = 62.011 \text{ kg}} \quad (\text{C.8.1})$$

For the rectangular cross-section, the minor second moment of area is

$$I = \frac{bh^3}{12} = \frac{(0.030)(0.004)^3}{12} = 1.600 \times 10^{-10} \text{ m}^4 = 160.0 \text{ mm}^4.$$

$$\boxed{I = 1.600 \times 10^{-10} \text{ m}^4 = 160.0 \text{ mm}^4} \quad (\text{C.8.2})$$

Multiplying the second moment by the elastic modulus gives the flexural stiffness,

$$EI = (200 \times 10^9)(1.600 \times 10^{-10}) = 32.00 \text{ N m}^2.$$

$$\boxed{EI = 32.00 \text{ N m}^2} \quad (\text{C.8.3})$$

The tip deflection of a cantilever under a concentrated tip load is $\Delta = PL^3/(3EI)$. Therefore, the equivalent minor-axis spring constant is

$$k = \frac{P}{\Delta} = \frac{3EI}{L^3} = \frac{3(32.00)}{(0.440)^3} = 1126.9722 \text{ N/m}.$$

$$\boxed{k = 1126.97 \text{ N/m}} \quad (\text{C.8.4})$$

Because the collision is short compared with the vibration and the projectile embeds in the initially stationary mass, momentum is conserved.

$$m_p v_p + M(0) = (M + m_p) \dot{u}_0.$$

Solving for the common velocity immediately after impact gives

$$\dot{u}_0 = \frac{m_p v_p}{M + m_p} = \frac{(0.011)(1020)}{62.011} = 0.1809356 \text{ m/s}.$$

$$\boxed{\dot{u}_0 = 0.181 \text{ m/s}, \quad u_0 = 0} \quad (\text{C.8.5})$$

The initial displacement is zero because the beam is vertical at impact. Following HW01 C.7 and course-note PDF page 252, the combined mass M_t contributes to both the inertia and the approximate gravity stiffness,

$$k_g = \frac{3M_t g}{2L} = 2073.8451 \text{ N/m}, \quad k_{\text{eff}} = k + k_g = 3200.8173 \text{ N/m}.$$

The natural frequency, cyclic frequency, and period are therefore

$$\omega_{n,\text{FBP}} = \sqrt{\frac{k}{M_t} + \frac{3g}{2L}} = 7.1844922 \text{ rad/s},$$

$$f_{n,\text{FBP}} = \frac{\omega_{n,\text{FBP}}}{2\pi}, \quad \tau_{\text{FBP}} = \frac{2\pi}{\omega_{n,\text{FBP}}} = 0.87454828 \text{ s}.$$

Full precision is retained in the calculations; final answers use the assignment's significant-digit convention.

$$\boxed{\omega_{n,\text{FBP}} = 7.18 \text{ rad/s}, \quad f_{n,\text{FBP}} = 1.143 \text{ Hz}, \quad \tau_{\text{FBP}} = 0.8745 \text{ s}} \quad (\text{C.8.6})$$

Substituting $u_0 = 0$ into the undamped initial-value solution gives

$$u(t) = \frac{\dot{u}_0}{\omega_{n,\text{FBP}}} \sin(\omega_{n,\text{FBP}} t).$$

The amplitude is $\dot{u}_0/\omega_{n,\text{FBP}} = 0.02518419 \text{ m}$. With guard digits retained in the phase,

$$\boxed{u(t) \approx 25.18419 \sin(7.1844922t) \text{ mm}}$$

$$u(18) = \frac{0.1809356}{7.1844922} \sin(7.1844922(18)) = -0.01241639 \text{ m}.$$

Therefore,

$$\boxed{u(18 \text{ s}) = -12.4 \text{ mm}} \quad (\text{C.8.7})$$

For twenty-three cycles, the duration is $t_{\text{end}} = 23\tau_{\text{FBP}} = 20.11461$ s. The first positive maximum occurs at

$$t_{\text{max}} = \frac{\pi}{2\omega_{n,\text{FBP}}} = 0.2186371 \text{ s}, \quad u_{\text{max}} = \frac{\dot{u}_0}{\omega_{n,\text{FBP}}} = 25.18419 \text{ mm}.$$

The two annotated markers identify this maximum and the response at eighteen seconds.

C8. Flex-beam pendulum using the course-note model

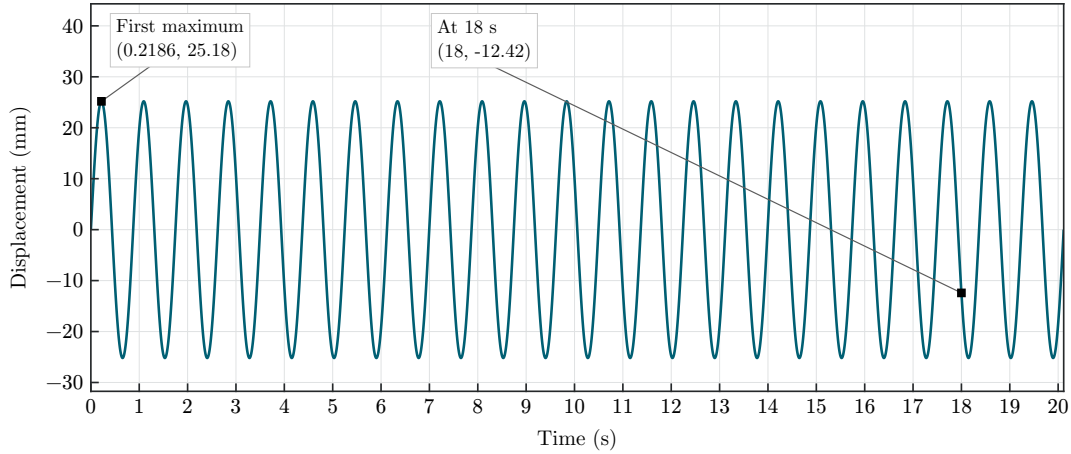


Figure 26: C.8. Twenty-three-cycle response using the course-note flex-beam-pendulum model.

$$N_{\text{osc}} = 23, \quad t_{\text{end}} = 20.11 \text{ s}, \quad u_{\text{max}} = 25.2 \text{ mm at } t = 0.219 \text{ s} \quad (\text{C.8.8.a})$$

$$u(18 \text{ s}) = -12.4 \text{ mm} \quad (\text{C.8.8.b})$$

Problem A.3 uses a simple pendulum; C.8 uses the course approximation combining cantilever bending and gravity. Their impact data and post-impact velocities differ, as do their frequencies.

Quantity	C.8 course model	A.3	Unit
$\dot{u}_0$	0.181	0.208	m/s
ω_n	7.18	4.43	rad/s
τ	0.8745	1.419	s
$ u _{\text{max}}$	25.2	46.9	mm
$u(18 \text{ s})$	-12.4	-43.5	mm

The course-model flex-beam pendulum has a 62.2% higher frequency and a 46.3% smaller amplitude than A.3. Both eighteen-second responses are negative, although their magnitudes differ. This comparison does not isolate bending stiffness, since the lengths, masses, and impact data also differ. The predicted amplitude is approximately $0.0572L$, so the small-motion assumptions remain a limitation.

$$\omega_{n,\text{FBP}} > \omega_{n,\text{A.3}}, \quad |u|_{\text{max,FBP}} < |u|_{\text{max,A.3}} \quad (\text{C.8.9})$$

Alternative model comparison.

The tension-dependent model described separately in HW01 C.7 gives $\omega_T = 6.6727$ rad/s, $\tau_T = 0.94163$ s, an amplitude of 27.1159 mm, and $u_T(18) = 18.0353$ mm. These are predictions of a different beam model. They are not used for the primary C.8 answers or plot above. The larger difference at eighteen seconds reflects the accumulated phase difference as well as the change in amplitude.

MATLAB code. *C.8 course-model response and calculated comparisons.*

```

01 % C.8. Flex-beam pendulum using course notes, PDF page 252.
02 cfg=hw01_settings();
03 L=.440; M=62; mp=.011; vp=1020; % m, kg, kg, m/s
04 E=200e9; h=.004; b=.030; g=9.81; % Pa, m, m, m/s^2
05 Nosc=23; t1=18; u0=0;
06 disp(table(L,M,mp,vp,E,h,b,g,Nosc,t1,u0));
07 I=b*h^3/12; EI=E*I; k=3*EI/L^3;
08 totalMass=M+mp; v0=mp*vp/totalMass;
09 % Tip slope/displacement = 3/(2L) in the course approximation.
10 slopePerDisplacement=3/(2*L);
11 gravityStiffness=totalMass*g*slopePerDisplacement;
12 effectiveStiffness=k+gravityStiffness;
13 wn=sqrt(effectiveStiffness/totalMass);
14 fn=wn/(2*pi); tau=2*pi/wn; amplitude=v0/wn;
15 eventT=[tau/4 t1];
16 t=hw01_grid(Nosc*tau,tau,eventT);
17 u=hw01_free(t,wn,0,u0,v0);
18 eventU=hw01_free(eventT,wn,0,u0,v0);
19 hw01_check_initial(wn,0,u0,v0);
20 assert(abs(totalMass*v0-mp*vp)<cfg.checkTolerance);
21 disp(table(I,EI,k,gravityStiffness,effectiveStiffness,v0,wn,fn,tau));
22 disp(table(eventT',eventU'*cfg.mmPerM, ...
23     'VariableNames',{'time_s','displacement_mm'}));
24 hw01_plot('C8',t,u*cfg.mmPerM,eventT,eventU*cfg.mmPerM,[], ...
25     'C8. Flex-beam pendulum using the course-note model');
26 % Recalculate A.3 from its own inputs for the requested comparison.
27 L_A3=.500; M_A3=45; mp_A3=.011; vp_A3=850;
28 wn_A3=sqrt(g/L_A3); v0_A3=mp_A3*vp_A3/(M_A3+mp_A3);
29 amplitude_A3=v0_A3/wn_A3;
30 fprintf('Frequency increase over A.3 = %.4g percent\n',100*(wn/wn_A3-1));
31 fprintf('Amplitude reduction from A.3 = %.4g percent\n', ...
32     100*(1-amplitude/amplitude_A3));
33 % Optional comparison only. The primary answer uses the course model.
34 preload=totalMass*g; lambda=sqrt(preload/EI);
35 kT=preload/(L-tanh(lambda*L)/lambda);
36 wnAlternative=sqrt(kT/totalMass);
37 tauAlternative=2*pi/wnAlternative;
38 amplitudeAlternative=v0/wnAlternative;
39 u10Alternative=hw01_free(t1,wnAlternative,0,u0,v0);
40 disp(table(wnAlternative,tauAlternative, ...
41     amplitudeAlternative*cfg.mmPerM,u10Alternative*cfg.mmPerM, ...
42     'VariableNames',{'alternative_wn_rad_s','alternative_period_s', ...
43     'alternative_amplitude_mm','alternative_u18_mm'}));

```

4 MATLAB Reproduction and Shared Functions

The runnable scripts in `matlab/` are the same files printed with each numerical solution. From that directory, run `run_all`. Each script displays its inputs and calculated results, checks its initial conditions, and updates the numerical data used by the Typst figures. The run also verifies the free responses against an independent matrix-exponential solution and checks final settling under changed inputs. Numerical calculations use SI units; conversion to display units occurs when producing tables and figures.

The source of the shared functions follows. Plot data are stored in `figures/src/solutions/data/`; run `make` in `figures/` to rebuild their PDFs. The MATLAB windows provide interactive data tips at the calculated event coordinates.

hw01_settings

```
01 function cfg = hw01_settings()
02 % Shared numerical and display settings. Calculations use SI units.
03 cfg.mmPerM = 1e3;
04 cfg.umPerM = 1e6;
05 cfg.secondsPerMinute = 60;
06 cfg.samplesPerCycle = 200;
07 cfg.samplesPerDecayTime = 50;
08 cfg.settlingFraction = 0.02;
09 cfg.checkTolerance = 1e-10;
10 cfg.colors = [0 95 115; 213 94 0; 37 40 42]/255;
11 cfg.dataDirectory = fullfile(fileparts(mfilename('fullpath')), ...
12     '..', 'figures', 'src', 'solutions', 'data');
13 end
```

hw01_check_initial

```
01 function hw01_check_initial(wn,zeta,u0,v0)
02 % Verify the specified displacement and velocity, before plotting.
03 cfg = hw01_settings();
04 [u,v] = hw01_free(0,wn,zeta,u0,v0);
05 assert(abs(u-u0)<=cfg.checkTolerance*max(1,abs(u0)));
06 assert(abs(v-v0)<=cfg.checkTolerance*max(1,abs(v0)));
07 end
```

hw01_free

```
01 function [u,v] = hw01_free(t,wn,zeta,u0,v0)
02 % Free response in metres and m/s, including all three damping cases.
03 assert(isscalar(wn) && wn>0 && isfinite(wn));
04 assert(isscalar(zeta) && zeta>=0 && isfinite(zeta));
05 if zeta < 1
06     alpha = zeta*wn;
07     wd = wn*sqrt(1-zeta^2);
08     B = (v0+alpha*u0)/wd;
09     harmonic = u0*cos(wd*t)+B*sin(wd*t);
10     u = exp(-alpha*t).*harmonic;
11     v = exp(-alpha*t).*(-alpha*harmonic ...
12         -u0*wd*sin(wd*t)+B*wd*cos(wd*t));
13 elseif zeta == 1
14     B = v0+wn*u0;
15     u = (u0+B*t).*exp(-wn*t);
16     v = (B-wn*(u0+B*t)).*exp(-wn*t);
17 else
18     q = sqrt(zeta^2-1);
19     sFast = -wn*(zeta+q);
20     sSlow = -wn/(zeta+q); % Avoid subtraction of nearly equal numbers.
21     Cfast = (v0-sSlow*u0)/(sFast-sSlow);
22     Cslow = (sFast*u0-v0)/(sFast-sSlow);
23     u = Cfast*exp(sFast*t)+Cslow*exp(sSlow*t);
24     v = sFast*Cfast*exp(sFast*t)+sSlow*Cslow*exp(sSlow*t);
25 end
26 end
```

hw01_settling

```

01 function [ts,us,tEnvelope] = hw01_settling(wn,zeta,u0,fraction)
02 % Final band entry for an underdamped release with ZERO initial velocity.
03 assert(wn>0 && zeta>0 && zeta<1 && u0~=0);
04 assert(fraction>0 && fraction<1);
05 wd = wn*sqrt(1-zeta^2);
06 alpha = zeta*wn;
07 % Extrema occur at n*pi/wd and have magnitude abs(u0)*exp(-alpha*t).
08 n = max(0,ceil(log(1/fraction)*wd/(alpha*pi))-1);
09 tPeak = n*pi/wd;
10 tZero = (n*pi+pi/2+atan2(alpha,wd))/wd;
11 relative = @(t) hw01_free(t,wn,zeta,1,0);
12 ts = fzero(@(t) abs(relative(t))-fraction,[tPeak tZero]);
13 us = hw01_free(ts,wn,zeta,u0,0);
14 tEnvelope = log(1/(fraction*sqrt(1-zeta^2)))/alpha;
15 assert(exp(-alpha*(n+1)*pi/wd)<=fraction*(1+1e-12));
16 end

```

hw01_grid

```

01 function t = hw01_grid(tEnd,period,eventTimes)
02 % Include exact event coordinates as well as samples per oscillation.
03 cfg = hw01_settings();
04 assert(tEnd>0 && period>0);
05 nIntervals = ceil(cfg.samplesPerCycle*tEnd/period);
06 t = unique([linspace(0,tEnd,nIntervals+1),eventTimes(:)']);
07 assert(all(t>=0 & t<=tEnd));
08 end

```

hw01_plot

```

01 function hw01_plot(name,x,Y,eventX,eventY,band,plotTitle)
02 % Refresh the existing Typst data and show equivalent MATLAB curves.
03 % Y and eventY use the display units specified by the figure's axis label.
04 cfg = hw01_settings();
05 path = fullfile(cfg.dataDirectory,[name '.json']);
06 d = jsondecode(fileread(path));
07 x = x(:)';
08 assert(size(Y,2)==numel(x) && all(isfinite(Y),'all'));
09 assert(size(Y,1)==numel(d.curves));
10 assert(numel(eventX)==numel(d.events));
11 d.x = x;
12 d.xlim = [x(1) x(end)];
13 d.band = band;
14 if nargin==7
15     d.title = plotTitle;
16 end
17 span = max(Y,[],'all')-min(Y,[],'all');
18 if span==0, span=max(1,abs(Y(1))); end
19 upperMargin = 0.13+0.25*~isempty(eventX);
20 d.ylim = [min(Y,[],'all')-0.13*span, ...
21     max(Y,[],'all')+upperMargin*span];
22 figure('Name',d.title,'Color','w');
23 hold on
24 styles = {'-','--','-.'};
25 for j=1:size(Y,1)
26     d.curves(j).y = Y(j,:);
27     plot(x,Y(j,:),styles{mod(j-1,numel(styles))+1}, ...
28         'Color',cfg.colors(mod(j-1,size(cfg.colors,1))+1,:), ...
29         'LineWidth',1.2,'DisplayName',d.curves(j).label);
30 end
31 if ~isempty(band)
32     yline(band,':k','HandleVisibility','off');
33     yline(-band,':k','HandleVisibility','off');
34 end
35 for j=1:numel(eventX)
36     d.events(j).x = eventX(j);
37     d.events(j).y = eventY(j);
38     d.events(j).value = sprintf('%.4g, %.4g',eventX(j),eventY(j));
39     point = plot(eventX(j),eventY(j),'ks','HandleVisibility','off');
40     datatip(point,eventX(j),eventY(j));
41 end
42 xlabel(d.xlabel); ylabel(d.ylabel); title(d.title);
43 grid on; box on; xlim(d.xlim); ylim(d.ylim);
44 if size(Y,1)>1, legend('Location','best'); end
45 fid = fopen(path,'w');
46 assert(fid~-1,'Cannot write figure data. ');
47 cleanup = onCleanup(@() fclose(fid));
48 d.curves = num2cell(d.curves);
49 d.events = num2cell(d.events);
50 payload = jsonencode(d);
51 % Typst represents an absent settling band as none, rather than an array.
52 payload = strrep(payload,'"band": []','"band":null');
53 fprintf(fid,'%s\n',payload);
54 end

```